1 Introduction to Statistical Learning

Statistical learning, as a whole, encompasses a vast set of tools/techniques used to help us understand data. They are classified into *supervised* or *unsupervised* tools. More generally, supervised statistical learning involves building a statistical model for predicting an output y based on one or more inputs. These problems are found everywhere: business, medicine, political science, and astrophysics! Unsupervised learning establishes a more intimate relationship between the inputs because there is no output y to be predicted. From unsupervised models, we can gather relationships and structures from said data. The models in these notes will fall into one of these two categories.

For example, a supervised model may look to predict whether a consumer participated in a fraudulent credit card transaction based on distance from home, whether the pin number has been used, or whether the order was purchased online.

Sometimes, no natural output variable is available. As such, we resort to unsupervised models. For instance, a dataset looks at the gene expressions for each of the cancer cell lines. Rather than having an output variable, we are interested in clustering the cell lines based on their gene measurements.

Statistical learning relies on general matrix algebra:

Definition 1.1 (Matrices and Vectors). **Matrices** can be thought of as a spreadsheet of numbers with n rows and p columns. Let $\mathbf{X}$ represent said matrix, then it contains np elements formatted as such:

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

In other words, we will use x_{ij} to represent the value of the j-th variable (column) for the i-th observation (row), where $i = 1, \dots, n$ and $j = 1, \dots, p$. If we are concerned with a column or row specifically, then we obtain a **vector**. Generally, we will rely on the columns of $\mathbf{X}$, or $\mathbf{x}_j$. The matrix can then be re-written as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_p \end{pmatrix}$$

The columns depict different features that are used to predict for find relations. For example, we could let $\mathbf{x}_1$ to represent the distance from home predictor in the credit card fraud example.

The **transpose** of a matrix swaps the rows with columns, or

$$\mathbf{X}^T = \begin{pmatrix} x_{11} & x_{21} & \dots & x_{n1} \\ x_{12} & x_{22} & \dots & x_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1p} & x_{2p} & \dots & x_{np} \end{pmatrix}$$

For a square matrix, the diagonal entries are the same between its default and transpose form.

As such, the columns of a transpose matrix is the rows of its original form:

$$\mathbf{X}^T = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_n \end{pmatrix}$$

In supervised learning models, when an output exists, the i-th output is denoted by y_i. We can store each of the n responses in an "output vector, or"

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Each observation, x_i, stores in values from each feature used to predict the response:

$$x_i = \begin{pmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{pmatrix}$$

Then $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ contains the set of each observation as well as its corresponding output.

In some cases, statistical models rely on the multiplication of either two matrices, a matrix and a vector, or two vectors. Matrix multiplication can be summarized by its requirements and outputs:

- Let **A** and **B** represent two matrices. In order for matrix multiplication to be possible, the number of columns in **A** must match the number of rows in **B**.
- Suppose the above assumption holds, or in other words, **A** and **B** are $q \times n$ and $n \times p$ matrices, respectively. The resulting matrix **AB** is a $q \times p$ matrix.

The (i, j)th element of $\mathbf{AB}$ is computed by multiplying each element of the ith row of $\mathbf{A}$ by the corresponding element of the jth column of $\mathbf{B}$ and then summing them together. For example, consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

Then,

$$\mathbf{AB} = \begin{pmatrix} (1)(5) + (2)(7) & (1)(6) + (2)(8) \\ (3)(5) + (4)(7) & (3)(6) + (4)(8) \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

1.1 Modeling Problems & Approaches

As a motivation, we consider a practical example. Suppose we are statistical consultants hired by a client to identify the association between renting costs and internal/external apartment properties in a certain city. In particular, we are tasked to identify how much the area (sqft), cost of in-unit amenities, distance to urban amenities (in miles), and presence of pools, fitness centers, and community events affect the annual rent of apartments.

If we can establish associations between apartment properties and rent, we can instruct our client to adjust prices according to what criteria are or are not satisfied, thereby maximizing landlord demand and profit.

In this setting, the apartment properties are the *input variables*, denoted by X, while the annual rent is an *output variable*, denoted by Y. In the scenario mentioned above, X_1 could represent the area, X_2 the cost of in-house amenities, X_3 as if whether a pool is provided or not, and so on. As for the output, Y is the projected annual rent.

More generally, suppose we observe a quantitative response Y and p different predictors, $X_1, X_2, \ldots, X_p$. We assume there exists some relationship between Y and $X = (X_1, X_2, \ldots, X_p)$ which can be written in the *general functional form*

$$Y = f(X) + \epsilon.$$

Here f is some fixed but unknown function of the predictors and ϵ is the error term, describing how close the predicted output is to its true, corresponding value. ϵ is independent of X and has a mean of zero. In context of statistical learning, f represents the *systematic* information that X provides about Y. Throughout these notes, we will employ different models in an attempt to best predict f.

1.2 Estimating f

All motivations behind statistical learning all preside on why and how we estimate f.

Each method for estimating f generally share certain characteristics. We always assume that we have observed a set of n different data points. These observations are called the

training data because we will use these observations to train, or teach, our method how to estimate f. The two reasons to estimate f lie in *prediction* and *inference*:

Prediction

In most situations, a set of inputs X are given, but the output Y cannot be easily obtained. However, because the error term averages to zero, we can predict Y using

$$\hat{Y} = \hat{f}(X)$$

where $\hat{f}$ is the "estimation" for f and $\hat{Y}$ is the resulting prediction for Y. $\hat{f}$ is often treated as a *black box* in the sense that we are not concerned of the form $\hat{f}$ takes as long as it provides accurate predictions for Y.

For example, suppose $X_1, \ldots, X_p$ are characteristics of a patient's blood sample that are easily measurable in a lab, and Y is a variable encoding the patient's risk for a severe adverse reaction to a particular drug. It makes sense to predict Y using X; we can then avoid giving the drug in question to patients who are at high risk of an adverse reaction—that is, patients for whom the estimate of Y is high.

The accuracy of $\hat{Y}$ relies on two error types: *reducible* and *irreducible*. $\hat{f}$ is almost never a perfect estimate for f; such accuracy is prone to error, namely reducible error because we can improve the accuracy of $\hat{f}$ by choosing the most appropriate statistical learning technique to estimate f. Recall that Y is also a function of ϵ, unable to be predicted by X. Therefore, the variability associated with ϵ is the irreducible error, because we cannot reduce the error introduced by ϵ regardless of how well we estimate f. From probability theory, we can show that the expected squared error

$$\begin{aligned} E[(Y - \hat{Y})^2] &= E[(f(X) + \epsilon - f(X))^2] \\ &= [f(X) - \hat{f}(X)]^2 + \mathrm{Var}(\epsilon) \end{aligned}$$

is a sum of the squared difference between predicted and actual Y (reducible) and the variance associated with ϵ (irreducible).

Inference

Compared to estimating $\hat{f}$, suppose we are interested in understanding the *association* between Y and $X_1, \ldots, X_p$. In this situation, we wish to estimate f, but not necessarily to make predictions for Y. Thus, we cannot $\hat{f}$ as a black box because we need to know its exact form. This can be obtained by asking ourselves the following questions:

- *Which predictors are associated with the response?* More often than not, a small fraction of the available predictors are substantially associated with Y. Identifying the handful of important predictors among a large selection can be extremely useful, depending on the application (ex. computational cost).

- *What is the relationship between the response and each predictor?* Some predictors may have a positive and/or stronger relationship with Y. That is to say, the larger a predictor becomes, the larger Y becomes. The resulting increase may be greater between different predictors, even if their relative increase is the same. Others may have the opposite effect. Depending on the complexity of f, the relationship between the response and a given predictor can also depend on the values of other predictors.
- *Can the relationship between Y and each predictor be adequately summarized using a linear equation, or is the relationship more complicated?* Historically, most methods for estimating f assumes some linear form, and is even the most desirable outcome in some situations. However, there are plenty of situations in which linear relationships are not the most appropriate choice. In fact, the true relationship is generally nonlinear; in such cases, a linear model may not provide the most accurate relationship between X and Y.

Prediction and inference can work separately or together in tandem. Depending on the goal, different methods in estimating f may be appropriate. For example, while linear models are easy to interpret and use, its performance falls off quickly if the true relationship is nonlinear.

We shift our focusing to answering: how do we estimate f? With the motivation behind us, we can explore many linear and nonlinear approaches for estimating f. Recall that our goal is to apply a statistical learning method to training data in order to estimate the unknown function f. More precisely, we want to find a function $\hat{f}$ such that $Y \approx \hat{f}(X)$ for any observation (X, Y). Generally speaking, most statistical learning methods for this task can be characterized as either *parametric* or *non-parametric*.

Parametric Methods

Parametric methods follow a two-step methodology:

1. Make an assumption about the functional form of f. For example,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p$$

   resembles a *linear model*. Instead of having to estimate an entirely arbitrary p-dimensional function $f(X)$, one only needs to estimate the $p + 1$ coefficients $\beta_0, \beta_1, \ldots, \beta_p$.

2. Choose a procedure to fit or train the model. In the case of the linear model, we want to minimize the squared error between the input and output values, otherwise known as (ordinary) least squares (to be discussed later). Apart from comparing $\hat{f}$ and Y, there are different ways to empirically fit and train a model.

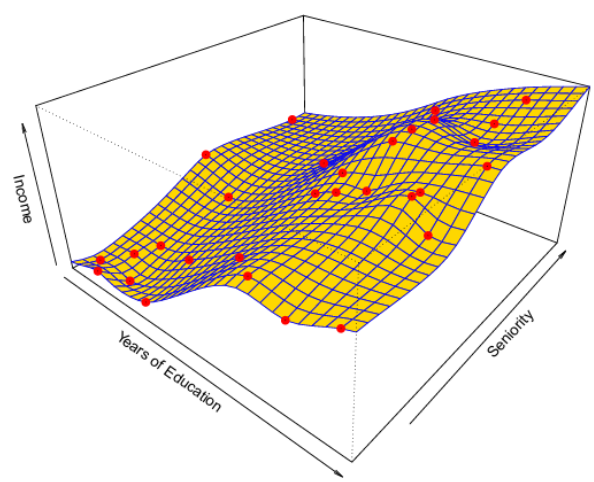
The model-based approached is described as *parametric* because it reduces the problem of estimating f down to one of estimating a set of parameters. Assuming a parametric form simplifies the problem of estimating f because it becomes much easier to estimate a set of parameters (such as coefficients) than it is to fit an entirely arbitrary function f. However, the disadvantage lies in that the model we choose will usually not match the true unknown form of f, and if it is too far, our estimate will be poor.

Solutions to this problem exist in the form of *flexible* models that can fit many different possible functional forms for f. But in general, fitting a more flexible model requires us to estimate a greater number of parameters, resulting in a phenomenon known as *overfitting* the data. In other words, the model follow the errors, or *noise*, too closely.

Non-Parametric Models

As opposed to parametric models, non-parametric models do not explicitly assume a functional form is possible and limits the need for finding parameters. Instead, they seek an estimate of f that gets as close to the data points as possible without being too rough/wiggly. By eliminating the constraint on f, non-parametric models have the potential to accurately fit a wider range of possible shapes for f.

A noticeable bottleneck from non-parametric models is that estimating f requires an extremely large number of observations in order to obtain an accurate estimate for f. Parametric models generally do not require as many observations, at least compared to non-parametric models, due to using a handful of parameters for estimating f.



A *thin-plate spline*, as shown above, is a non-parametric method that uses interpolation to perfectly passes through all of the training points. By perfectly fitting training data, the complex structure of f will be highly sensitive to and therefore overfit test data. Splines can be "smoothed," which allows for less abrupt changes in curvature and direction,

generally used to estimate trends from noisy training data as opposed to perfectly fitting it.

1.3 Types of Statistical Learning Problems

We compare two sets of machine learning problems: Supervised vs. Unsupervised Learning and Regression vs. Classification.

1.3.1 Supervised vs. Unsupervised Learning

Every example so far has fallen within the realm of supervised learning. For each observation of the predictor measurement(s) x_i, $i = 1, \ldots, n$, there is an associated measurement y_i. Supervised learning is referred to as *learning with training*. Given X, y, we want to fit the mapping between X and y with training data and make predictions with the new y given new X in the test dataset. The ultimate goal is to minimize the distance between $\hat{y}$, the predicted value, and y.

By contrast, unsupervised learning presents a more challenging situation. We still have a vector of measurements x_i, but no response y_i this time. This situation is referred to as unsupervised because we are *learning without training*, or without the supervision of a response variable. Under unsupervised learning, we want to identify trends or patterns within our observations. For instance, *clustering analysis* is an unsupervised method used to group observations into distinct groups.

1.3.2 Regression vs. Classification

Regression and classification problems are distinguished by the form its response variable assumes. Typically, variables take one of two forms:

- **Quantitative**: These variables take on numerical values. Examples include one's income, age, height, etc...
- **Qualitative (Categorical)**: Take on values from one of K classes, or categories. Examples include one's marital status (married/not married), red blood cell types (A/B/AB/O), or the brand of product purchased (A/B/C).

Regression problems have a quantitative response variable and *classification* problems have a qualitative response variable. Here are examples of each:

- Linear regression models are almost always regression problems. For instance, depending on the shape of the data, the annual rent model from Section 1.1 can be expressed through a linear regression model because an apartment's rent is quantitative. One of its possible predictors, the presence of external amenities such as a pool or gym, would be qualitative (ex. 0 for no pool, 1 for pool).

- Logistic regression models output either "0" or "1," making it a great choice for classification problems. They are often classified as *binary classification* problems because it outputs two possible values. Therefore, for responses with two possible outcomes, logistic models are a very popular choice. For instance, logistic regression can be used for disease diagnosis, spam detection, or whether borrowers default on a loan.
- Some flexible models can be used for both regression and classification. Two such models are K-nearest neighbors and decision trees, which will be introduced later.

There are extensions to classification problems, beyond the binary output that a logistic regression model assumes.

- Multi-classification problems have a response variable with more than 2 classes (identifying between a dog, cat, or bird).
- Multi-label classification problems have outputs that can belong to more than one class. Typical classification problems assume one class assignment, whereas multi-label classification can have be assigned one, multiple, or even none of the response outputs. For example, political and sports can be an output for one newspaper, whereas politics and finance can be an output for a different newspaper.

1.4 Accuracy, Bias-Variance Tradeoff

1.4.1 Accuracy

In order to evaluate the performance of a given data set, we need some way to measure how well its predictions actually match the observed data. That is to say, we need to quantify the extent to which the predicted response value for a given observation is close to the true response value for that observation.

In a regression setting, the most commonly used metric is the *mean squared error* (MSE), given by

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$$

where $\hat{f}(x_i)$ is the prediction that $\hat{f}$ gives for the ith observation. The MSE will be small if the predicted responses are very close to the true response, and large if the predicted and true responses differ substantially for some of the observations. Because the MSE is computed using the training data, we refer to it as the *training MSE*. Instead, we are interested in the accuracy of the previous predictions that we obtain when we apply our method to previously unseen test data. Suppose we are interested in developing an algorithm to predict a patient's diabetes diagnosis based on clinical measurements

(weight, blood pressure, height, age, family history of disease). We are not very interested in whether or not the method accurately predicts diabetes risk for patients used to train the model, since we already know which of those patients have diabetes. Therefore, the results from the training data are used on *future patients*.

Mathematically speaking, suppose we fit our statistical learning method on our training observations $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ and we obtain the estimate $\hat{f}$. Then, we compute $\hat{f}(x_1), \hat{f}(x_2), \ldots, \hat{f}(x_n)$. We are not interested in whether $\hat{f}(x_1) \approx y_i$; instead, we want to know whether $\hat{f}(x_0)$ is approximately equal to y_0, where (x_0, y_0) is a previously unseen test observation not used to train the statistical learning method.

The ultimate goal is to select the model for which the *test MSE* is as smallest as possible. Unfortunately, there is no guarantee that the method with the lowest training MSE will also have the lowest test MSE. This is because many statistical methods specifically estimate coefficients so as to minimize the training set MSE. While the training MSE is typically small in these cases, the test MSE if often much larger.

Accuracy in classification problems takes on a less complicated form. In short, accuracy is measured by the number of correct predictions in each of the K classes over the total number of predictions.

$$\text{Accuracy} = \text{Average}\left(\mathbb{I}(y_i = \hat{y}_i)\right)$$

where $\mathbb{I}(y_i = \hat{y}_i)$ is an *indicator variable* that equals 1 if $y_i = \hat{y}_i$ and 0 if $y_i \neq \hat{y}_i$. Likewise, the *test error rate* associated with a set of test observations of the form (x_0, y_0) is given by

$$\text{Average}\left(\mathbb{I}(y_i \neq \hat{y}_i)\right)$$

The *training error rate*, or the proportion of mistakes that are made if we apply our estimate $\hat{f}$ to the training observations, is computed accordingly:

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}(y_i \neq \hat{y}_i)$$

While its computation is easier compared to regression problems, accuracy is not always the best metric.

Suppose a medical dataset is interested in predicted whether a patient has cancer through common symptoms: fatigue, pain, depression (which are measured in some capacity), in which a logistic regression model is fit to the training data. The training and test data have a 98% no cancer and 2% cancer split. A 98% accuracy on the test data seems promising, but it could also imply that it correctly predicts roughly 99% of the non-cancer patients but only 50% of the cancer patients. This results in a lot of false negatives on patients with cancer, resulting from an extreme *class imbalance*. In such cases, we must ensure that the minority class has high accuracy. This is typically more

difficult, because the minority class has less data to train from, making it harder to establish a boundary between each response class.

1.4.2 Bias-Variance Tradeoff

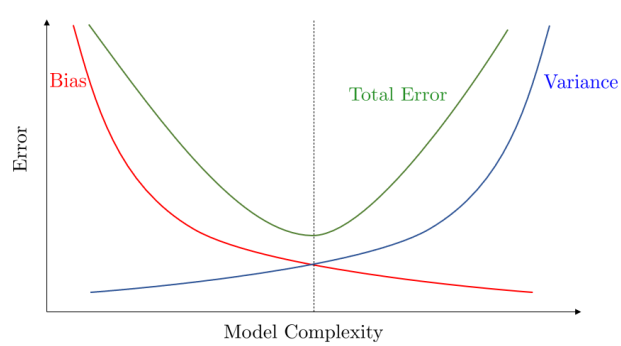
Model accuracy in a regression setting is contingent on two competing properties of statistical learning methods. We describe each in detail:

- **Bias** is the error introduced by making assumptions to simplify the learning process. For example, using a linear model to a dataset with a nonlinear relationship is will have high bias, for it is trying to simplify but does not fit the model. This is *underfitting*.
- **Variance** is the error caused by small fluctuations in the model's test data. A model with high variance will not only capture the patterns in the training data, but the noise as well (overfitting). High variance models work well with training data but poorly on new test data.

Recalling the irreducible variance of the error terms ϵ, the *expected test MSE* (not regular MSE!) is decomposed into the sum of three fundamental quantities: the variance of $\hat{f}(x_0)$, the squared bias of $\hat{f}(x_0)$, and the variance of the error terms ϵ. That is,

$$E\left[(y_0 - \hat{f}(x_0))^2\right] = \text{Var}(\hat{f}(x_0)) + \left[\text{Bias}(\hat{f}(x_0))\right]^2 + \text{Var}(\epsilon)$$

Bias and variance act in opposite directions as we increase the model complexity. By using a more complex model, we make more restrictions, therefore relying on fewer assumptions and lowering the bias. However, complex models are harder to fit, thereby increasing the variance. If we simplify the model, more assumptions are made which increases the bias and decreases the variance. This phenomenon is commonly referred to as the **bias-variance tradeoff**.



Making a model too simple or complex will make the test MSE large, which is why we want to typically meet in the middle. In the case of the diagram above, the minimal total error is achieved when the bias and variance intersect. This should make sense intuitively: balancing bias and variance eliminates the results of making simplifying assumptions or a large list of parameters.

Nonetheless, the task of finding a method for which both the variance and squared bias are low is a challenging one. Moreover, in real-life situations where f is unobserved, test MSEs are not explicitly obtainable. However, bias-variance tradeoff is important to keep in mind when designing statistical models.

1.5 Resampling Methods

The process we have streamlined thus far has been to build a model, train it on existing data, and then test it on new data. We also discussed the difference between the *training error rate* and the *test error rate*. The test error is the average error that results from using a statistical learning method to predict the response on a new observation—that is, a measurement that was not used in training the method. The test error is readily available if such test data is available. Unfortunately, this is not usually the case.

A common approach to this solution is *holding out* a subset of the training observations from the fitting process, and then applying the statistical learning method to those held out observations. This genre of approaches is an example of *resampling*, or repeatedly drawing samples from a training dataset to improve model accuracy and evaluate performance. There are multiple approaches to resampling, each carrying their own strengths and weaknesses.

1.5.1 Training Set vs. Test Set

Each resampling method attempts to estimate the test error associated with fitting a particular statistical learning method on a set of observations. The *training set vs. test set*, or the *validation set* approach, is a very simple strategy for this task. In involves randomly dividing the available set of observations into two parts, a *training set* and a *validation set* or *hold-out set*. The model is fit on the training set, and the fitted model is used to predict the responses for the observations in the validation set. The error is assessed using MSE for regression models and a misclassification rate for classification models.

In most cases, the split for train and validation sets are 50%-75% train and 25%-50% validation, depending on the size of the dataset or if a classification model has a class imbalance.

The validation set approach is conceptually simple and easy to implement. However, it has two potential drawbacks.

1. The validation estimate of the test error rate can be highly variable, depending on precisely which observations are included in the training set and which observations are included in the validation set, leading to some dependence of randomness for a low test error rate.

2. Only a subset of the observations, as opposed to the entire data, are used to fit the model. Since statistical methods tend to perform worse when trained on fewer observations, this suggests the validation set error rate may tend to *overestimate* the test error rate for the model fit on the entire data set.

The following two sections will refine the training vs. test set approach that addresses these two issues.

1.5.2 Leave One One Cross-Validation

Leave-one-out cross validation (LOOCV) is closely related to the training vs. test set approach of section 1.5.1, but it attempts to address that method's drawbacks.

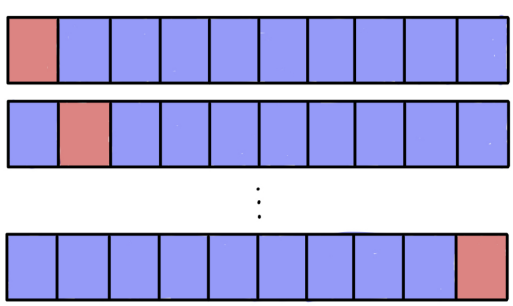
Like the training vs. test set approach, LOOCV involves splitting the set of observations into two parts. Given n observations, the LOOCV method constructs n *folds*, or samples, each containing $n-1$ training data. The remaining observation is the test data. The accuracy is measured by averaging the performance of the n models. In a regression model, n MSEs are computed, with the LOOCV estimate being the average of those n test error estimates.

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{MSE}_i$$

In a classification setting, the LOOCV error rate takes the form

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{Err}_i$$

where $\mathrm{Err}_i = \mathbb{I}(y_i \neq \hat{y}_i)$.



The blue rectangles indicate the set of data we are training, and the red rectangle is the singular test data. There are some major advantages to using LOOCV:

- It has far less bias. The training set contains $n-1$ observations, therefore utilizing almost the entire data set for fitting the model. In contrast, the training set vs. test set approach generally only trains between 50% and 75% of the full data set.
- There is no randomness compared to the training set vs. test set approach. LOOCV will output the same result each time.

Because we are running n models, LOOCV has the potential to be expensive to implement, making it very time consuming if n is large or if the model itself is complex or slow to fit. LOOCV is a very general method and can be used with any kind of predictive modeling.

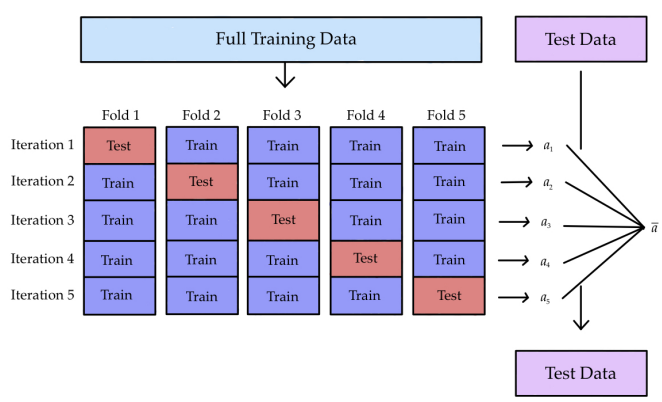
1.5.3 k-fold Validation

An alternative to LOOCV is *k-fold cross-validation (CV)*. It relaxes the restrictions imposed on partitioning data. Instead of generating n fixed models, we randomly partition the training data into $k < n$ folds. For each fold, we train on $k-1$ folds and test on the remaining $k-1$ folds. This procedure is repeated k times, resulting in k estimates of the test error, $\text{MSE}_1, \text{MSE}_2, \ldots, \text{MSE}_k$. The k-fold CV estimate is computed by averaging these values,

$$\text{CV}_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \text{MSE}_i.$$

The same treatment is given to classification problems:

$$\text{CV}_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \text{Err}_k.$$



In a typical k-fold CV setup, $k = 5$ (as shown in the last page) or $k = 10$ provides enough unique models while keeping the computational cost low, which is a great advantage if n is extremely large or if the fitting process is lengthy.

By definition, LOOCV is a special case of k-fold validation where $k = n$.

Putting computational cost aside, it is worth noting that k-fold CV often gives more accurate estimates of the test error rate than does LOOCV. The Bias-Variance Tradeoff discussed previously plays a major role here.

Recall that the training vs. test set approach can lead to overestimates of the test error rate because the training set used to fit the statistical learning method contains only 50%-75% of the entire data set. With this logic, the LOOCV will give approximately unbiased estimates of the test error, since each training set contains $n - 1$ observations. Performing k-fold CV (say $k = 5$ or $k = 10$) will lead to an intermediate level of bias, since each training set contains approximately $\frac{n(k-1)}{k}$ observations, fewer than the LOOCV approach, but significantly more than the training vs. test set approach. Therefore, in terms of reducing the bias, LOOCV is preferred to k-fold CV.

As for variance, LOOCV has higher variance than k-fold CV with $k < n$. This is because we are effectively averaging the output of n fitted models, each of which is trained on an almost identical set of observations, making the outputs highly correlated with each other. In k-fold CV with $k < n$, we are averaging k fitted models that are less correlated with each other because there is less overlap between models. Because the mean of many highly correlated quantities has higher variance than does the mean of loosely correlated quantities, the resulting test error estimate from LOOCV tends to have higher variance than does the test error estimate resulting from k-fold CV.

In summary, the bias-variance tradeoff in k-fold cross-validation is associated with the choice of k. Therefore, under these considerations, choosing $k = 5$ or $k = 10$ empirically yields test error estimates that suffer from neither excessively high bias nor high variance.

1.6 Examples

Example 1.2 (ISLP 2.4.1). For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer.

(a) The sample size n is extremely large, and the number of predictors p is small.

We would most likely expect a flexible method to perform better as it is more capable of capturing the large amount of data we are given. An inflexible method may cause to larger residual error among all of the data as it can only take on a fixed form that may not accurately predict the true relationship.

(b) The number of predictors p is extremely large, and the number of observations is small.

The inflexible method would perform better. With a large amount of predictors and few observations, flexible models would more likely overfit the data and generate misleading predictions. With an inflexible model only have a few parameters to optimize (ex. coefficients in linear regression) which lowers the possibility of overfitting.

(c) The relationship between the predictors and response is highly non-linear.

Flexible methods are far more suitable for finding non-linear, complex relationships as they do not follow a fixed form, unlike inflexible methods. As a result, flexible methods have higher degrees of freedom, or number of data points that can vary.

(d) The variance of the error terms, i.e. $\sigma^2 = \text{Var}(\epsilon)$, is extremely high.

Inflexible methods are better tailored when the variance of the noise is high. Flexible methods will try to fit the noise and only lead to a larger increase in variance.

Example 1.3 (ISLP 2.4.5). What are the advantages and disadvantages of a very flexible (versus a less flexible) approach for regression or classification? Under what circumstances might a more flexible approach be preferred to a less flexible approach? When might a less flexible approach be preferred?

Answer: Flexible models can capture more complex and non-linear relationships better than a less flexible method. However, it is more subjective to overfitting than less flexible models. They would be preferred in situations in which the data set is large enough and the underlying relationship is certainly non-linear. A less flexible method would be preferred when the sample size is small and the underlying relationship appears to be linear.

Example 1.4 (ISLP 2.4.6). Describe the differences between a parametric and a non-parametric statistical learning approach. What are the advantages of a parametric approach to regression or classification (as opposed to a non-parametric approach)? What are its disadvantages?

Answer: A parametric model assumes f follows a functional form and tries to estimate its parameters using the data. Non-parametric methods, on the other hand, do not make assumptions about f and instead seek an estimate of f that gets as close to the data points as possible.

Parametric models are typically less computationally intensive and are easier to estimate parameters instead of an arbitrary f. Parametric models, if poorly chosen, can have too much bias and lead to an undesirable error.

Example 1.5 (Multiple Choice). Given training error and test error values for various models, which model would we expect to be most accurate on new, unseen data?

(a) The model for which the train error and test error are closest to each other.

(b) The model for which the test error is lowest.

(c) The model for which the train error is lowest.

(d) The model for which the flexibility is greatest.

Solution: B. The test error measures how well the model performs on unseen data. Therefore, the model with the lowest test error is expected to generalize best to new observations.

Example 1.6 (Multiple Choice). Imagine we have a training data set with 500 observations. Which of the following could be a disadvantage of using 500-fold cross-validation, or LOOCV.

(a) The training process could be inaccurate, due to only using one observation for training each step.

(b) The training process could be slow, due to it needing to be repeated 500 times.

(c) LOOCV can only be used with classification problems.

(d) LOOCV can only be used with regression problems.

Solution: B. LOOCV has no restriction on problem type, eliminating (c) and (d). Fitting 500 models and running them on singular test data can substantially increase the computational cost, making (b) the best option. The training process uses 499 points, using almost all of the data. This results in low bias and typically a more reliable estimate of the test error, which is a benefit of LOOCV, therefore ruling out (a).

2 K-Nearest Neighbors

2.1 Methodology

After learning the basis of statistical methods, we can apply them to practice in commonly known models. Of which, we will briefly dive into K-nearest neighbors, which can be performed in both a regression and classification setting.

Before proceeding, we introduce a definition that applies to many statistical methods:

Definition 2.1 (Hyperparameters). In a statistical learning method, **hyperparameters** are "settings" applied to more freely control the learning process and model structure. Hyperparameters are not learned directly from the data, but instead configured to optimize the model's learning process. Finding the best hyperparameter or combination of them is known as **tuning**.

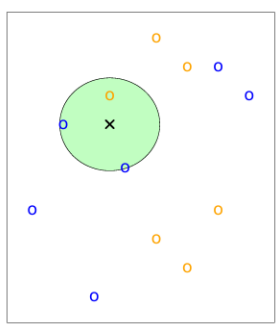
For example, when we discussed k-fold CV in Section 1.5.3, k would be an example of a hyperparameter because it controls the format in which error is summarized (number of iterations and errors to be averaged).

K**-nearest neighbors** follows a "similar inputs have similar outputs" motto. The method varies on whether we are looking at a regression or classification problem.

- *Classification*: For a test input $\mathbf{x}$, assign the most common label amongst its K most similar training inputs, or closest neighbors
- *Regression*: For a test input $\mathbf{x}$, output the average of its K closest neighbors.

These "closest neighbors" are computed using various distance metrics. Euclidean distance is typically preferred on low-dimensional data, but can become undesirable with a significant amount of features. In this section, we will default to Euclidean distance for simplicity.

To motivate the idea, we begin by understanding KNN classification problems.



Taken from *An Introduction to Statistical Learning: Applications in Python*

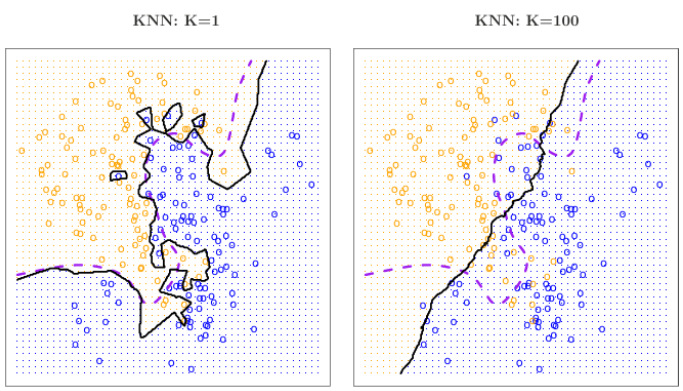
In the figure seen above, the test input is marked by a cross. The green region encompasses the three closest observations ($k = 3$), either marked by the blue or gold class. Between the 3 nearest observations, two of them are from the blue class, signaling the majority and resulting in a blue test output.

In the event of a tie (only possible if k is even) between classes, common methods for handling them include randomly selecting one class, picking the class of the closest observation, or decreasing k by one.

Definition 2.2 (Decision Boundaries). In n-dimensional data, decision boundaries separate classes from each other for classification problems. It is a line (in 2D), a plane (in 3D), a hyperplane (in 4D), and so forth.

Decision boundaries basically represent the threshold at which the model switches its prediction from one class to another. For example, in a credit fraud data set, an observation beyond one side of a decision boundary would predict "fraud detected" and an observation beyond the other would predict "fraud not detected." If a point lies exactly on the decision boundary, the model is undecided, as each class has a 0.5 probability of occurring. Decision boundaries can be linear or nonlinear, depending on the model used to fit the data and hyperparameters. To summarize, decision boundaries are what define classification problems.

K is the hyperparameter used in this method and has a drastic effect on the KNN classifier obtained.



Taken from *An Introduction to Statistical Learning: Applications in Python*

As illustrated by the figure:

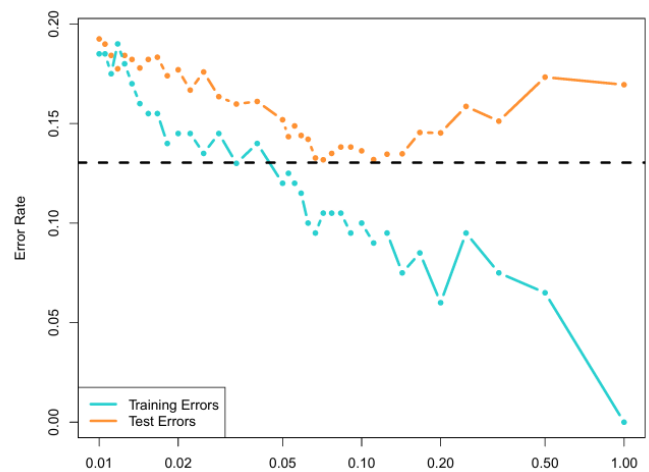
- When $K = 1$, the decision boundary is too flexible. The model makes decisions based on a single nearest neighbor, making it highly sensitive noise and small changes in the training data, a strong signal of overfitting. Because every training

point is its own neighbor, the model perfectly "memorizes" the training set, resulting in 0% error on that data. This is a low bias, high variance characteristic for this model.

- As K grows, the method becomes less flexible and produces a decision boundary that is closely linear. This is a high bias, low variance characteristic for this model.

$K = 1$ and $K = 100$ are both extreme on this data set; neither give good predictions and would result in undesirable test error rates. Thus, there is not a strong relationship between training and test error rates.

The same applies to KNN regression problems.



Taken from *An Introduction to Statistical Learning: Applications in Python*

With $K = 1$, the KNN training error rate is 0, but the test error rate may be quite high. Generally, as we use more flexible classification methods, the training error rate will decline but the test error rate may not. As shown in the figure above, the KNN test and training errors are plotted as a function of $\frac{1}{K}$. As $\frac{1}{K}$ increases, the method becomes more flexible. For regression models, the training error rate consistently declines as the flexibility increases. The test error, however, has a U-shape, declining at first (roughly at $K = 10$) before increasing again when the method becomes overly flexible and overfits.

The K-nearest neighbors model is just one application of the bias-variance tradeoff, and makes model optimization a challenging task.

2.2 Examples

Example 2.3 (ISLP 2.4.7). The table below provides a training data set containing six observations, three predictors, and one qualitative response variable.

Obs	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

Suppose we wish to use this data set to make a prediction for Y when $X_1 = X_2 = X_3 = 0$ using K-nearest neighbors.

(a) Compute the Euclidean distance between each observation and the test point, $X_1 = X_2 = X_3 = 0$.

In order from observations 1-6: $3, 2, \sqrt{10}, \sqrt{5}, \sqrt{2}, \sqrt{3}$

(b) What is our prediction with $K = 1$? Why?

Green. The point closest to $(0, 0, 0)$ is observation 5 of distance $\sqrt{2}$, whose corresponding Y is green.

(c) What is our prediction with $K = 3$? Why?

Red. The three closest points to $(0, 0, 0)$ are observations 2, 5, and 6 of respective distances 2, $\sqrt{2}$, and $\sqrt{3}$. Their corresponding Y's are red, green, and red respectively, giving red the majority vote and therefore the resulting output.

(d) If the true decision boundary in this problem is highly non-linear, then would we expect the *best* value for K to be large or small? Why?

We would expect K to be small because smaller values of K result in more flexibility which are better for non-linear decision boundaries.

Example 2.4. Imagine we want to perform KNN regression with $K = 2$ using a training dataset with just four observations and just one predictor. The training data consists of the four points

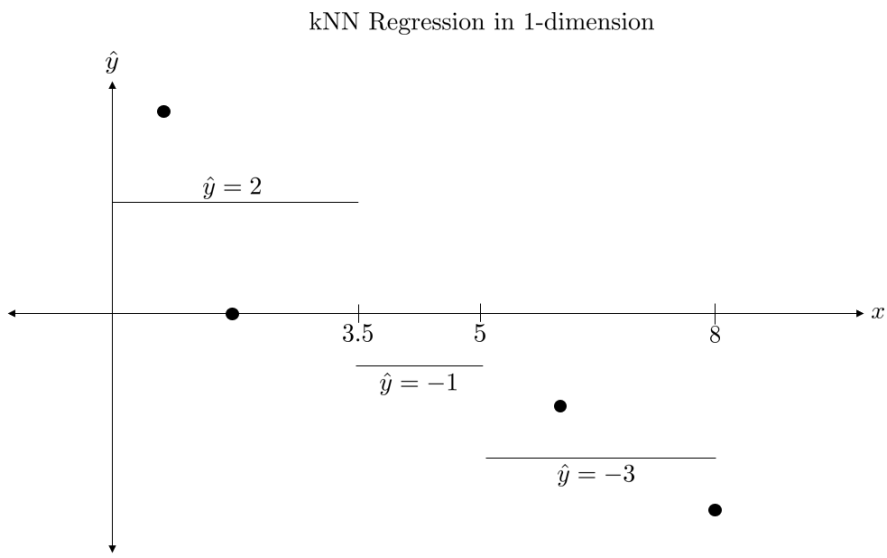
$$\{(1, 4), (2, 0), (6, -2), (8, -4)\}$$

(i.e. the input 1 corresponds to the output 4). Sketch a graph of what the corresponding prediction function will look like. Do not worry about boundary points (when an input is equally far from two different points)

- When $x < 3.5$, the two closest neighbors are $x = 1$ and $x = 2$. Therefore, we take the average of their y's, which is 2.

- When $3.5 < x < 5$, the two closest neighbors are $x = 2$ and $x = 6$. The average of their y's is $\hat{y} = -1$.

- When $x > 5$, the two closest neighbors are $x = 6$ and $x = 8$. The average of their y's is $\hat{y} = -3$



Example 2.5. We perform KNN regression with $K = 50$ using a training set with 10000 observations.

(a) For a given input $\mathbf{x}$, how many distances must be calculated to produce a predicted output corresponding to the input $\mathbf{x}$?

While we are looking for the 50 nearest neighbors, they cannot be decided until all 10000 distances are computed.

(b) If we next want to make a prediction for a new input $\mathbf{x}_0$, do we need to calculate more distances? If so, how many? If not, why not?

The first test input does not get "added" into the model because it is not part of the training data. Therefore, another 10000 distances (not 10001!) are required to see which 50 observations are closest to this new output.

3 Linear Regression

Linear regression is a very simple approach for supervised learning and predicting a quantitative response. While linear regression models themselves are quite dull and lacking complexity, it is still widely used and serves as an important basis before tackling more challenging statistical models.

We will familiarize ourselves with the *Advertising* data set before proceeding.

- The *Advertising* data set consists of the *sales* of that product in 200 different markets, along with advertising budgets for the product in each of those markets for three different media: *TV*, *radio*, and *newspaper*. These are the features, or input variables, each expressed in thousands of dollars.
- The output variable is *sales*, or the number of units of the product sold.

Shown below is a plot of sales vs. each of the advertising methods (input variables).

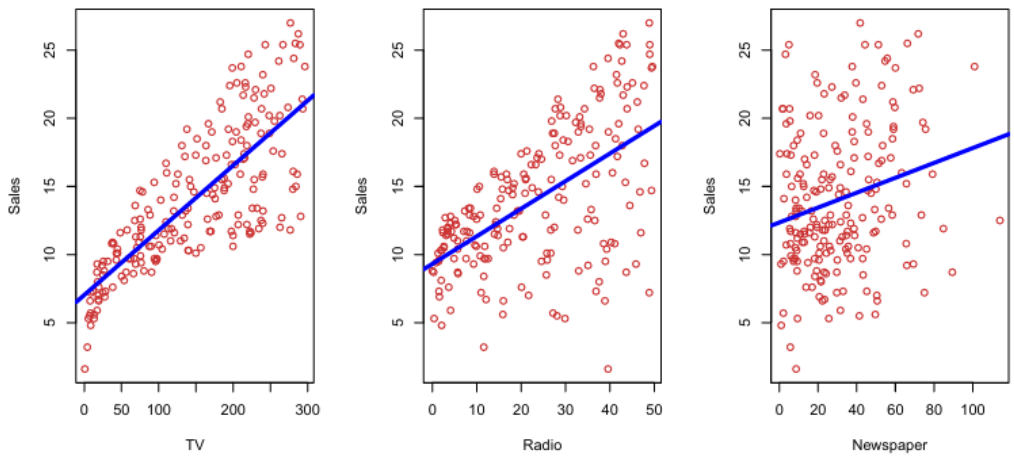


Figure 3.1: Taken from *An Introduction to Statistical Learning: Applications in Python*

In this section, we wish to develop a "line of best fit" that combines all three features to predict the number of sales (one plot). Suppose we are asked to suggest, on the basis of this data, a marketing plan for next year that will result in high product sales. What information would be useful in order to provide such a recommendation? Here are some questions we want to address:

1. *Is there a relationship between advertising and budget sales?* If the data cannot provide evidence of an association between advertising and expenditure and sales, then one might argue that no money should be spent on advertising!

2. *How strong is the relationship between advertising budget and sales?* If a relationship exists, how strong is the relationship? Does knowledge of the advertising budget provide a lot of information about product sales?

3. *Which media are associated with sales?* It is possible that not all three media—TV, radio, and newspaper—are associated with media. To determine this, we must find a way to separate out the individual contributions of each medium to sales when we have spent money on all three media.

4. *How large is the association between medium and sales?* If we increase amount spent on advertising by a dollar, by what amount will the sales increase? How accurately can we predict this amount of increase?

5. *How accurately can we predict future sales?* For any given level of television, radio, or newspaper advertising, what is our prediction for sales, and what is the accuracy of this prediction?

6. *Is the relationship linear?* If there is roughly a straight-line relationship between advertising expenditure (in the various media) and sales, then linear regression is an appropriate tool.

7. *Is there synergy among the advertising media?* For example, spending $50,000 on television advertising and $50,000 on radio advertising is associated with higher sales than allocating $100,000 to either television or radio individually. In marketing terms, this is a *synergy effect*, while in statistics it is called an *interaction effect*.

A linear regression model can answer each of these questions! These questions will be answered separately, then contextually applied to the *Advertising* data set.

3.1 Simple Linear Regression

3.1.1 Model Introduction

Simple linear regression is the most straightforward approach for predicting a quantitative response Y on the basis of a single predictor variable X. It assumes that there is a linear, or close to linear, relationship between X and Y, making this a parametric model. Mathematically, we can describe the relationship as

$$Y \approx \beta_0 + \beta_1 X$$

...or we are regressing Y on X. For example Y represents sales and X may represent one of TV, radio, or media. Let's say we regress sales onto TV, then

$$Y = \beta_0 + \beta_1 \times \text{TV}$$

This is common slope-intercept form: β_0 is the intercept term and β_1 is the slope coefficient for TV. After using training data to estimate $\hat{\beta}_0$ and $\hat{\beta}_1$, we can predict future sales using

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \times \text{TV}$$

where $\hat{y}$ indicates a prediction of Y on the basis of $X = x$. The β terms are the *coefficients* or *parameters* to be estimated.

3.1.2 Coefficient Estimation and Accuracy

Let $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ represent n observation pairs, each containing a measurement of X and a measurement of Y. Our goal is to estimate coefficient estimates $\hat{\beta}_0, \hat{\beta}_1$ such that the linear model fits the data well. In other words, so that

$$y_i \approx \hat{\beta}_0 + \hat{\beta}_1 x_i \quad \text{for } i = 1, \ldots, n.$$

This means, we want $\hat{\beta}_0$ and $\hat{\beta}_1$ such that the resulting line is as close as possible to the n data points (in this case, $n = 200$ for the data set). We can measure *closeness* in a variety of ways, but here we want to minimize the *least squares* criterion.

Definition 3.1 (Residual Sum of Squares (RSS)). Suppose $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be the prediction for Y based on the ith value of X. Then $e_i = y_i - \hat{y}_i$ represents the ith *residual*, or the difference between the ith observed value and the ith response value predicted by the linear. The **residual sum of squares** is

$$\text{RSS} = e_1^2 + e_2^2 + \cdots + e_n^2$$

In context of a simple linear regression model,

$$\text{RSS} = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \ldots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2$$

Calculus can quickly prove that the minimizers are

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2} \qquad \hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}}$$

where $\overline{y} = \frac{1}{n}\sum_{i=1}^n y_i$ and $\overline{x} = \frac{1}{n}\sum_{i=1}^n x_i$ are the sample means. The boxed coefficients define the *least squares coefficient estimates* for simple linear regression.

Using the *Advertising* data, the least squares fit for the regression of sales onto TV reveals $\hat{\beta}_0 = 7.03$ and $\hat{\beta}_1 = 0.0475$. That is to say, according to this approximation, an additional \$1,000 spent on TV advertising is associated with selling roughly 47.5 additional units of the product. The line of best fit can be observed in the figure on the next page.

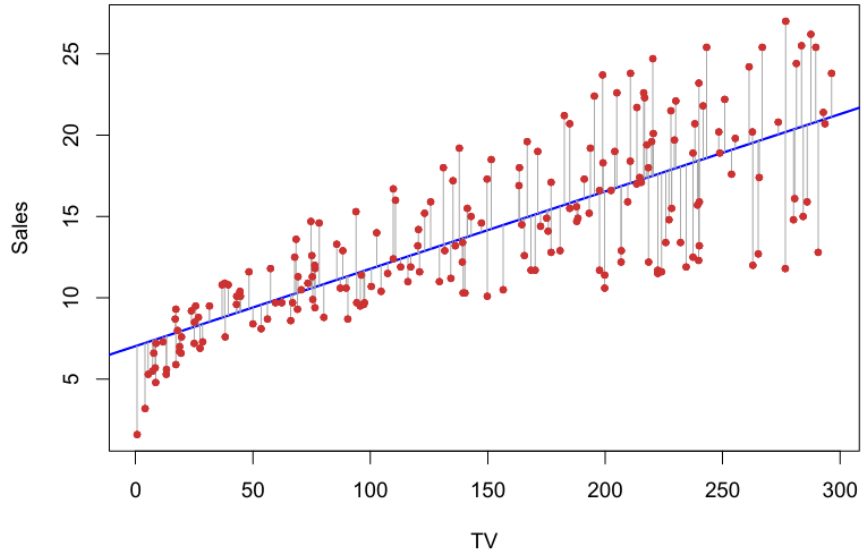


Figure 3.2: Taken from *An Introduction to Statistical Learning: Applications in Python*

If we plotted a 3D graph showing the RSS as a function of β_0 and β_1, there is some (β_0, β_1) that would minimize the RSS, thereby making it a convex relationship.

These coefficient estimates are almost never 100% accurate. Recall from earlier that we assume the *true* relationship between X and Y takes the form $Y = f(X) + \epsilon$ for some unknown f, where ϵ is the zero-mean random error term. In a simple linear regression, this relationship becomes

$$Y = \beta_0 + \beta_1 X + \epsilon.$$

We typically assume the error is completely independent of X. If $\epsilon = 0$, then we say the model is *deterministic*. However, we assume $\epsilon \sim \mathcal{N}(0, \sigma^2)$, or in other words, that it follows a normal distribution.

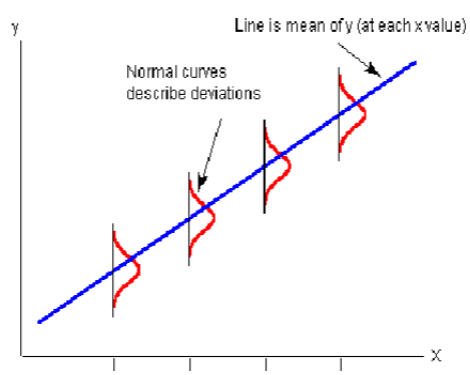


Figure 3.3

y can range from the values shown on the vertical lines, and the expected value of y, or the mean, is what the blue line passes through.

To motivate this idea, suppose we have a simple linear regression model of the form

$$Y = 2 + 3X + \epsilon,$$

generating 100 random X's corresponding Y's.

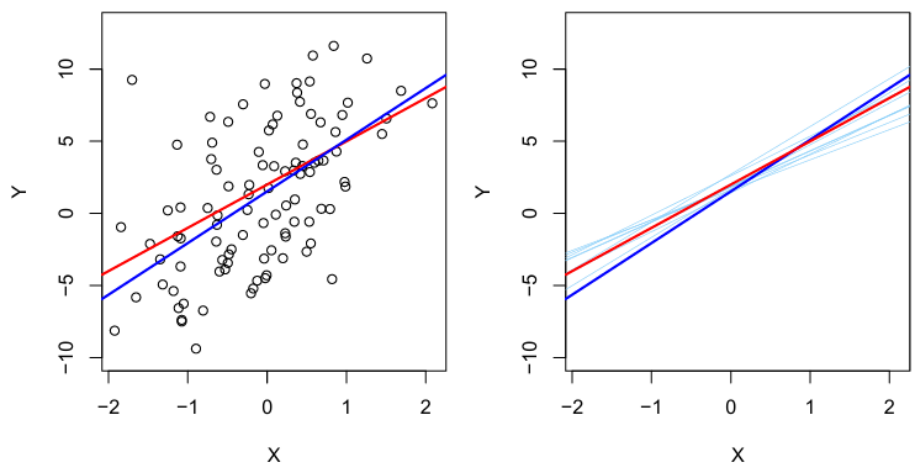


Figure 3.4: Taken from *An Introduction to Statistical Learning: Applications in Python*

On the left-hand graph, the red line is the true relationship (population regression line) whereas the blue line is the least squares estimate (sample regression line). We have access to a set of observations from which the least squares line is obtainable, but the population regression line is unobserved.

On the right-hand graph, ten different data sets from the same model Y were generated and their corresponding least squares lines are plotted. We only have one data set, and so what does it mean that two different lines describe the relationship between the predictor and the response? The concept of these two lines is a natural extension of the standard statistical approach of using information from a sample to estimate characteristics of a large population.

Suppose we are interested in knowing the population mean μ of some random variable Y. μ is unknown, but we have access to n observations from Y, $y_1, \ldots, y_n$, which can be used to estimate μ. A reasonable estimate is $\hat{\mu} = \overline{y}$, where y is the sample mean. While the sample and population mean are different, the sample mean will provide a good estimate of the population mean. Using $\hat{\mu}$ to estimate μ is *unbiased*, because $\hat{\mu}$ may over or underestimate μ on different sets of observations. These unbiasedness also applies to the approximation of regression coefficients β_0 and β_1. On the basis of a particular data

set, our estimates won't be exactly equal to β_0 and β_1, but if we averaged the estimates over a large number of data sets, then the average of these estimates will be exact!

Next, we are interested in the quantitatively computing the difference between $\hat{\mu}$ and μ.

Definition 3.2 (Standard Error and Variance). The **standard error** and **variance** of the sample mean $\hat{\mu}$ is given by

$$\text{Var}(\hat{\mu}) = [\text{SE}(\hat{\mu})]^2 = \frac{\sigma^2}{\mu}$$

where σ is the standard deviation of each of the realizations y_i of Y. The standard error tells us the average amount that the estimate $\hat{\mu}$ differs from the actual value of μ.

The equation reveals how the deviation shrinks with $n-$the more observations we have, the smaller the standard error of $\hat{\mu}$. In the same vein, we can measure how close $\hat{\beta}_0$ and $\hat{\beta}_1$ are to the true values of β_0 and β_1. We apply the following formulas:

$$\boxed{[\text{SE}(\hat{\beta}_0)]^2 = \sigma^2 \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^n (x_i - \overline{x})^2}\right] \qquad [\text{SE}(\hat{\beta}_1)]^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \overline{x})^2}}$$

where $\sigma^2 = \text{Var}(\epsilon)$.

For these formulas to be strictly valid, we must assume that the errors ϵ_i for each observation have common variance σ^2 and are uncorrelated. This does not necessarily apply to the *Advertising* data set (see Figure 3.2), but the formula is a good approximation nonetheless. Note that from both formulas above, the coefficient standard error approximations are smaller when the x_i are more spread out; intuitively, we have more *leverage* to estimate a slope in this case. Also, $\text{SE}(\hat{\beta}_0) = \text{SE}(\hat{\mu})$ if $\overline{x} = 0$ (because $\hat{\beta}_0 = \overline{y}$).

Generally, σ^2 is unknown, but can be estimated from the data. This estimate of σ is the *residual standard error*, given by

$$\boxed{\text{RSE} = \sqrt{\frac{\text{RSS}}{n-2}}.}$$

Standard errors can be used to compute confidence intervals.

Definition 3.3 (Confidence Intervals (CI)). A k% **confidence interval (CI)** is defined as a range of values such that with k% probability, the range will contain the true unknown value of the parameter. If we were to repeat this procedure a very large number of times and constructed the confidence interval for each sample, then k% of the intervals computed will contain the true parameter value.

The most commonly used percentage is 95%. That is to say, if we repeated samples and construct the confidence interval for each sample, 95% of the intervals will contain the

true unknown value of the parameter. For linear regression, the 95% CI for β_1 approximately takes the form

$$\hat{\beta}_1 \pm 2 \cdot \text{SE}(\hat{\beta}_1)$$

As shown, the range is defined in terms of lower and upper limits computed from the sample of data. A general statistical conclusion would say, "there is approximately a 95% chance the interval

$$\left[\hat{\beta}_1 - 2 \cdot \text{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \text{SE}(\hat{\beta}_1)\right]$$

will contain the true value of β_1." Similarly, the 95% confidence interval for β_0 is

$$\hat{\beta}_0 \pm 2 \cdot \text{SE}(\hat{\beta}_0)$$

Going back to the *Advertising* data,

- The 95% CI for β_0 is [6.130, 7.935]. That is to say, in the absence of advertising, we are 95% confident that the true sales amount will fall somewhere between 6,130 and 7,935 units. Or, we could say that sales, on average, will fall somewhere between 6,130 and 7,935 units.
- The 95% CI For β_1 is [0.042, 0.053]. That is to say, for each \$1,000 increase in television advertising, we are 95% confident that the true increase in sales is between 42 and 53 units. Or, we could say that sales, on average, will increase between 42 and 53 units.

Standard errors are also used to conduct *hypothesis tests* on the coefficients. We describe each in detail:

1. *Null hypothesis* (H_0): Claims "there is no relationship between X and Y, or mathematically $\beta_1 = 0$. In other words, the model is better off with only the intercept term β_0.
2. *Alternative hypothesis* (H_a): Claims "there is some relationship between X and Y, or mathematically $\beta_1 \neq 0$. In other words, the model is better off with the intercept term β_0 and feature coefficient β_1.

How can we measure whether we should reject the null or alternative hypothesis? First, we need to determine whether $\hat{\beta}_1$, our estimate for β_1, is sufficiently far from zero that we can be confident that β_1 is also non-zero. How far is far enough? It depends on the accuracy of $\hat{\beta}_1$, or $\text{SE}(\hat{\beta}_1)$. If $\text{SE}(\hat{\beta}_1)$ is large, than $\hat{\beta}_1$ must be large enough to offset this difference and therefore reject the null hypothesis.

Definition 3.4 (t-statistic). The **t-statistic** is the ratio of the difference in a number's estimated value from its assumed value to its standard error.

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)}$$

The t-statistic measures the number of standard deviations that $\hat{\beta}_1$ is away from 0. If no such relationship between X and Y exists, then we expect that the t-statistic formula follows a t-distribution with $n-2$ degrees of freedom. The t-distribution, for $n > 30$, is quite similar to the standard normal distribution.

Assuming $\beta_1 = 0$, we wish to compute the probability of observing any number equal to $|t|$ or larger in absolute value. This probability is known as the p-value. Roughly speaking, we interpret the p-value as follows:

A small p-value indicates that it is unlikely to observe such a substantial association between the predictor and the response due to chance, in the absence of any real association between the predictor and the response.

Therefore, for a small p-value, we can infer that there is an association between the predictor and the response and *reject the null hypothesis*. That is, we conclude a relationship between X and Y exists. Common cutoffs for the p-value include 0.05 and 0.01, with corresponding t-statistics of around 2 and 2.75, respectively (for $n = 30$).

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	7.0325	0.4578	15.36	< 0.0001
`TV`	0.0475	0.0027	17.67	< 0.0001

Table 3.5: Taken from *An Introduction to Statistical Learning: Applications in Python*

Applying simple linear regression on the *Advertising* data, we have that the coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ are substantially larger than their respective standard errors, resulting in a moderate t-statistic and low p-values. We would reject the null hypothesis and conclude that $\beta_0 \neq 0$, $\beta_1 \neq 0$. In context of *Advertising*, a positive relationship exists between advertising expenditures through TV and the number of units sold. Therefore, we can say a \$1,000 increase in TV advertising expenditure would increase the number of sales by 47.5 units.

3.1.3 Assessing Accuracy of the Model

Once the null hypothesis has been rejected in favor of the alternative hypothesis, we also want to quantify the extent to which the model fits the data. The quality of a linear regression fit is generally assessed on two measurements: the residual standard error and the R^2 statistic.

Definition 3.5 (Residual Standard Error (RSE)). The **residual standard error (RSE)** is an estimate of the standard deviation of the unknown error ϵ. More precisely, it is the average amount that the response will deviate from the true regression line. If n is the number of observations in the data, then

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{n-2}} = \sqrt{\frac{1}{n-2} \cdot \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

The residual standard error is considered a measure of the lack of error of the model to the data. If the predictions obtained using the model are very close to the true outcome values, or $\hat{y}_i \approx \hat{y}$ for $i = 1, \ldots, n$, then the RSE will be small and we can conclude the model fits the data well. We can expect the opposite conclusion should the predicted and observed values are far off.

Definition 3.6 (R^2 Statistic). The R^2 **Statistic**, or the *coefficient of determination*, estimates how much of the variation in Y is explained by X. This quantity is a ratio, commonly expressed as a percentage. It uses the formula

$$R^2 = \frac{SSTO - RSS}{SSTO} = 1 - \frac{RSS}{SSTO}$$

where $SSTO$ is the **sum of squared total**, computed using

$$SSTO = \sum_{i=1}^{n} (y_i - \overline{y})^2$$

A more concrete explanation for $SSTO$ is the amount of variability inherent in the response before regression is applied. In contrast, RSS measures the amount of variability that is left unexplained after performing the regression, thereby making $SSTO - RSS$ the amount of variability in the response explained by performing the regression. Dividing by $SSTO$, we get the proportion mentioned in Def. 3.6.

If R^2 is closer to 1, then the regression model explains a large proportion of the variability in the response. If R^2 is closer to 0, then the regression model does not explain much of the variability; this can occur if the linear model is wrong, the error variance σ^2 is high, or both.

Quantity	Value
Residual standard error	3.26
R^2	0.612
F-statistic	312.1

Table 3.6: Taken from *An Introduction to Statistical Learning: Applications in Python*

Table 3.6 shows these quantities applied to the *Advertising* data,

1. The RSE is 3.26, implying that actual sales in each market deviate from the true regression line by approximately 3,260 units, on average. That is to say, even if the model were correct and the true values of the unknown coefficients β_0 and β_1 were known exactly, any prediction of sales would be still be off by about 3,260 units on average. The mean value of *sales* over all markets is approximately 14,000 units (ISLP 78), and so the percentage error is $\frac{3260}{14000} = 23\%$.
2. We have $R^2 = 0.612$. That is to say, 61.2% of the variability in *sales* is explained by a linear regression on TV.

The R^2 statistic has an interpretability advantage over the RSE because it always lies between 0 and 1. It can still be challenging to determine what a "good" R^2 value is and it depends on the application. Some applications may see more extraneous factors that do not make it into the model, but still contribute to the variability. In those cases, low R^2 values can even be realistic.

Definition 3.7 (Correlation). **Correlation** is another measure of linear relationships between X and Y. It is defined as

$$r = \mathrm{Cor}(X, Y) = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \overline{y})^2}}$$

R^2 and $\mathrm{Cor}(X, Y)$ are both measures of linear relationships; in fact, one can prove that $r^2 = R^2$ in a simple linear regression setting (r^2 is the squared correlation here). Correlation does not extend as easily in multiple linear regression as it quantifies the relationship between two pairs of variables; R^2 will apply quite nicely, however.

3.2 Multiple Linear Regression

While nice when applicable, simple linear regression is, in practice, not incredibly useful. We generally want to establish an association between the response Y and multiple predictor variables. As in the *Advertising* data, we have three advertising mediums: TV, newspaper, and media. We only looked at the relationship between TV and sales; a standard linear regression model would include all three as features (at least to start out with). The question stands: how can we extend our analysis of the *Advertising* data to accomodate two additional predictors (newspaper and media)?

We could run three separate simple linear regression models, using a different advertising medium as a predictor. This method is not entirely satisfactory, however. First off, it is unclear how to make a single prediction of sales given three advertising media budgets when each of the budgets are associated with their own regression

model. Second, each of the three regression models do not take the other two budgets into consideration in forming estimates for the regression coefficients.

Instead of fitting a separate simple linear regression model for each predictor, a better approach is to extend the simple linear regression model so that it can directly accommodate multiple predictors. All we need to do is to add a separate slope coefficient for each predictor. More generally, suppose we have p distinct predictors. Then, the multiple linear regression model takes the functional form

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon,$$

where X_j represents the jth predictor and β_j quantifies the association between that variable and the response. We interpret β_j as the **average effect** on Y of a one unit increase in X_j, **holding all other predictors fixed**. Using the *Advertising* example, we can establish the multiple linear regression model

$$\text{sales } = \beta_0 + \beta_1 \text{TV } + \beta_2 \text{radio } + \beta_3 \text{newspaper } + \epsilon$$

3.2.1 Regression Coefficient Estimation

Coefficient estimation for multiple linear regression follows an identical approach to simple linear regression, but the calculations will not be as easy to work with. Again, we want to make estimates for each regression coefficient and we can make predictions with the formula

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p.$$

We aim to minimize the sum of squared residuals

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \cdots - \hat{\beta}_p x_{ip})^2,$$

where $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ being the least squares coefficients. Instead of 2 coefficients to optimize in simple linear regression, we have $p+1$ to optimize. The computations to minimize the RSS combine both calculus and linear algebra.

Let $\mathbf{X}$ be the $n \times (p+1)$ matrix containing every observation as well as their corresponding value for each predictor (the leftmost column contains the intercept terms, so each entry is 1), and $\mathbf{y}$ be the $n \times 1$ vector containing the associated response for each observation. The coefficient estimation $(p+1) \times 1$ vector β that minimizes the RSS uses the computation

$$\beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.$$

β contains the optimized coefficients for the intercept and each predictor. The equation used is known as the **normal equation**. To keep things simple, the normal equation uses

orthogonal projections to find a linear combination of $p+1$ vectors such that the result is closest to $\mathbf{y}$ as possible in that subspace. If we performed the matrix multiplication, the dimensionality on both sides will match. Generally, $\mathbf{X}^T\mathbf{X}$ is invertible, but there exist cases in which the normal equation fails.

1. One or more features are dependent on each other. For example, $x_3 = 2x_1 + x_2$.
2. The amount of predictors exceed the number of observations ($p+1 > n$).

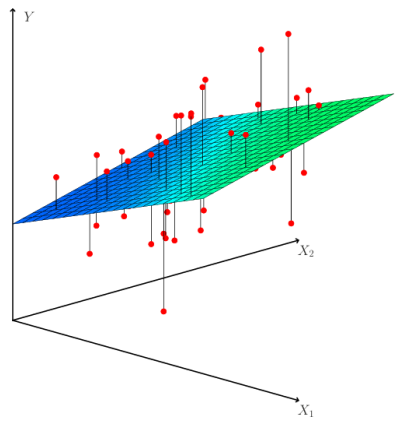


Figure 3.7: Taken from *An Introduction to Statistical Learning: Applications in Python*

Shown above is an example of linear regression with 2 predictors, X_1 and X_2. The resulting model is a plane of best fit.

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	2.939	0.3119	9.42	< 0.0001
`TV`	0.046	0.0014	32.81	< 0.0001
`radio`	0.189	0.0086	21.89	< 0.0001
`newspaper`	-0.001	0.0059	-0.18	0.8599

Table 3.8: Taken from *An Introduction to Statistical Learning: Applications in Python*

When multiple linear regression is applied to the *Advertising* data set, one can make the following observations:

1. For a \$1,000 increase in spending on TV advertising, the amount of sales is expected to increase by approximately 46 units, holding the other features constant.
2. For a \$1,000 increase in spending on radio advertising, the amount of sales is expected to increase by 189 units, holding the other features constant.
3. For a \$1,000 increase in spending on newspaper advertising, the amount of sales is expected to decrease by 1 unit, holding the other features constant.

Although not shown, the coefficient for newspaper in the simple linear regression model was significantly non-zero, but close to zero in the multiple linear regression. The p-value is quite large, standing around 0.86. This stems from the fact that in the simple regression case, the slope term represents the average increase in product sales associated with a $1,000 increase in newspaper advertising, ignoring other predictors such as TV and radio. One could conclude this result as

At the 0.05 significance level, newspaper has a p-value of 0.86, thereby signaling that it is not a strong predictor of sales when TV and radio are held fixed.

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

Table 3.9: Taken from *An Introduction to Statistical Learning: Applications in Python*

The results from the multiple regression model does not imply *newspaper* is a completely uncorrelated feature, however. Looking at the correlation matrix (Table 3.9), the correlation between newspaper and sales is 0.35, indicating that markets with high newspaper advertising tend to have high radio advertising. Assuming the regression model is correct, spending more on radio advertising is linked to higher sales. But, by the correlation between radio and newspaper, we expect a similar level of expenditure on newspaper advertising even if it is not directly associated with sales. This is a situation in which newspaper acts as a "surrogate" for radio advertising; newspaper gets "credit" for the association between radio on sales.

This situation reflects what is referred to as **confounders**, or **confounding variables**. A confounder

1. affects the response; and
2. is associated with one or more predictors.

Radio is a perfect example of a confounding variable here. The regression model in Table 3.8 suggests its positive relationship with *sales* but it also correlated with newspaper, as observed in Table 3.9.

Another example of a confounding variable would be running a regression of shark attacks vs. ice cream sales at a beach. Higher temperature would result in more people visiting the beach, and in turn, increase the number of shark attacks. Ice cream sales is a confounding variable; it can establish associations with external factors such as temperature and indirectly affect the amount of shark attacks.

3.2.2 Common Questions

This section seeks to address and answer four problems concerning multiple linear regression.

Is there a relationship between the response and predictors?

Recall that in simple linear regression, we only had to check if $\beta_1 = 0$ to determine whether there is a relationship between the response and the predictor. In a multiple logistic regression setting with p predictors, we need to ask whether all of the coefficients are zero. That is to say,

- Null hypothesis H_0: $\beta_1 = \beta_2 = \cdots = \beta_p = 0$
- Alternative hypothesis H_a: At least one of $\beta_j \neq 0$ for $j = 1, 2, \ldots, p$.

This hypothesis test is conducted through the F-statistic.

Definition 3.8 (F-statistic).

$$F = \frac{\frac{SSTO - RSS}{p}}{\frac{RSS}{n-p-1}}$$

where $SSTO = \sum_{i=1}^{n}(y_i - \overline{y})^2$ and $RSS = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$.

If the linear model assumption holds, then

$$E\left[\frac{RSS}{n-p-1}\right] = \sigma^2,$$

and provided H_0 is true,

$$E\left[\frac{SSTO - RSS}{p}\right] = \sigma^2$$

Therefore, if no relationship exists between predictors and the response, we would expect the F-statistic to be close to 1. On the other hand, if H_a holds, then

$$E\left[\frac{SSTO - RSS}{p}\right] > \sigma^2,$$

so we expect F to be greater than 1.

Quantity	Value
Residual standard error	1.69
R^2	0.897
F-statistic	570

Table 3.10: Taken from *An Introduction to Statistical Learning: Applications in Python*

In the *Advertising* data, the F-statistic is very far from one, we are inclined to reject the null hypothesis H_0 and suggest that at least one of TV, radio, and newspaper has a relationship with sales. This is reinforced by very low p-values in the intercept, TV, and radio (Table 3.8).

What if the F-statistic was closer to 1? How far from 1 does the F-statistic need to be before we can reject H_0 and conclude that a relationship exists? They usually depend on n and p! When n is large, an F-statistic close to 1 might still suggest evidence against H_0. However, if n is small, the F-statistic needs to be extremely far from 1.

If H_0 holds and ϵ_i follows a normal distribution, then the F-statistic follows an F-distribution.

We may want to test if only a subset q ($q < p$) of the coefficients are zero. The new hypothesis test becomes:

$$H_0 : \beta_{p-q+1} = \beta_{p-q+2} = \cdots = \beta_p = 0,$$

where we omit the variables at the end of the list. In this case, we fit a second model that uses all the variables *except* those last q. Suppose the residual sum of squares for that model is RSS_0. Then the resulting F-statistic is

$$F = \frac{\frac{RSS_0 - RSS}{q}}{\frac{RSS}{n-p-1}}$$

From Table 3.8, a t-statistic and p-value are also given. These provide information about whether each individual predictor is related to the response, after adjusting for the other predictors. As is turns out, this is equal to the F-test that omits that single predictor from the model, leaving all others in ($q = 1$). It reports the partial effect of adding that variable to the model. This is evidenced by the confounding variable situation with the radio and newspaper predictors.

Why do we concern ourselves with the F-statistic if we can look at the individual p-values? This rationale does not apply well in general, but especially when p is large (the room for more confounding variables grow).

Consider an example where $p = 100$ and we conclude $\beta_1 = \beta_2 = \cdots = \beta_{100} = 0$, so no variable is truly associated with the response. However, 5% of the p-values associated with each variable will be < 0.05 by chance. In other words, we expect to see approximately five *small* p-values even in the absence of any true association between the predictors and response, which in fact, we will likely observe at least one p-value below 0.05 by chance! Therefore, we will potentially make incorrect conclusions by going off either the t-statistic and p-values alone.

Recall that the F-statistic adjusts for the number of predictors (see Def. 3.8). Hence, if H_0 is true, there is only a 5% chance the F-statistic will result in a p-value below 0.05,

regardless of the number of predictors and observations. Generally, using a F-statistic works when p is relatively small and small compared to n. If p is large, we resort to different methods, as we will see in the next question.

Which features are truly important?

Typically, if the number of predictors is small, we can infer from p-values and the t-statistic. However, we use a different methodology if we think the response is associated with only a subset of the predictors. The process of determining which predictors are associated with a response, in order to fit a single model involving only those predictors, is referred to as *variable selection*.

Suppose $p = 2$. We can consider 4 models: (1) A model without X_1, X_2, (2) A model with only X_1, (3) A model with only X_2, and (4) A model with both X_1 and X_2. We can then select the best model out of all the models we have considered. To determine which model is best, we can use different statistics. These include *Mallow's* C_p, *Akaike information criterion* (AIC), *Bayesian information criterion* (BIC), and *adjusted* R^2. These will be discussed in the next chapter.

Unfortunately, there are 2^p subsets of predictors one can make. So, trying out every subset on even a moderate amount of predictors makes this approach unfeasible. To find the optimal subset of predictors, we consider three classical tasks:

1. *Forward Selection*: Start with the *null model*, or only containing the intercept. Fit p simple linear-regressions and add to the null model the variable that results in the lowest RSS. Add the variable that results in the lowest RSS. Then, add the model the variable that results in the lowest RSS for the new two-variable model. Continue until some stopping threshold is met.

2. *Backward Selection*: Start with the *full model*, or the intercept plus each predictor. Remove the variable with the largest p-value, or the variable that is the least statistically significant. Fit the new $(p-1)$-variable model, and the variable with the largest p-value is removed. Continue until some stopping threshold is met.

3. *Mixed selection*: Also known as stepwise building. This is a combination of forward and backward selection. Start with the null model and perform forward selection. However, if at any point, a threshold is exceeded with one predictor, remove it from the model. Continue forward and backward selection until all variables are below the threshold, and all variables outside the model would be statistically insignificant.

The common threshold is measuring the predictor's p-value (ex. remove any predictors whose corresponding p-value is greater than 0.05) when included in the model.

Backward selection cannot be applied if $p > n$, while forward selection is always usable. Forward selection is a "greedy" approach in the sense that variable included early on

may become redundant later when new ones are added. Mixed selection can be a remedy to this.

How do we measure the model's fit?

As mentioned in simple linear regression, two of the most common numerical measures of model fit are the RSE and R^2 (fraction of variance explained). They are computed in the same fashion as for simple linear regression.

R^2-Statistic

Recall that for simple linear regression,

$$R^2 = [\text{Cor}(X, Y)]^2$$

In multiple linear regression, it turns out that $R^2 = \left[\text{Cor}(Y, \hat{Y})\right]^2$. One property of the fitted linear model is that it maximizes this correlation among all possible linear models.

Table 3.10 revealed a R^2 of 0.8972 on the *Advertising* full linear regression model. A model including only TV and media has a R^2 of 0.89719. It turns out that adding new predictors will always increase R^2, even if they have a weak association with the response. This is because adding another variable always results in a decrease in the RSS on the training data, although not necessarily on the test data. Because R^2 is computed on the training data, it must increase.

If R^2 increases by a marginally small amount when features are added, it provides additional evidence that they are not needed in the model. Essentially, they provide no real improvement in the model fit to the training observations and its conclusion will likely lead to poor results on independent test samples from overfitting.

The model containing only TV as a predictor had an R^2 of 0.61 (Table 3.6). Adding radio to the model leads to a significant improvement in R^2, implying that a model that uses TV and radio expenditures to predict sales is substantially better than one that uses only TV advertising. This can be further verified by checking the p-value of radio in a model containing only TV and radio as predictors.

Residual Standard Error

The model containing only TV and radio as predictors has an RSE of 1.681 and the mdoel that also contains newspaper has an RSE of 1.69 (Table 3.10). On the contrary, the model containing only TV has an RSE of 3.26. This further suggests that a model using TV and radio expenditures is much more accurate (on the training data) than one that only uses TV spending. Furthermore, given that TV and radio are in the model, there is no point in adding newspaper as a predictor. How can RSE increase when the RSS decreases as new

features are added? The RSE is defined as

$$\boxed{\text{RSE} = \sqrt{\frac{RSS}{n-p-1}}.}$$

We can observe that this RSE formula will reduce to its simple regression equivalent by setting $p = 1$. While the RSS can increase, including more variables can have higher RSE if the decrease is RSS is insignificant relative to the increase in p.

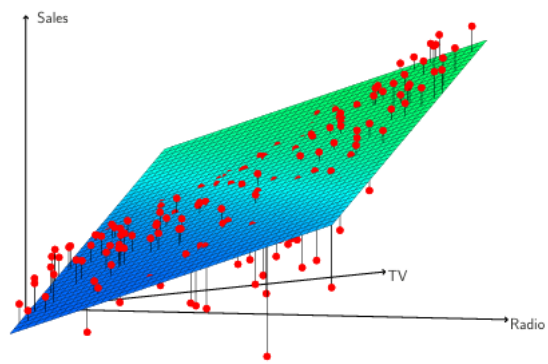


Figure 3.11: Taken from *An Introduction to Statistical Learning: Applications in Python*

It can also be useful to plot the data and can reveal problems with a model that are not visible from numerical statistics. Figure 3.11 shows a 3D-plot of TV and radio vs. sales. In particular,

- The linear model seems to overestimate sales for instances in which most of the advertising money was spent exclusively on TV or radio.
- The linear model seems to underestimate sales for instances where the budget was split between the two media.

This non-linear pattern suggests *synergy* or an *interaction effect* between the advertising media, where combining the media together results in a bigger boost to sales than any single medium.

How do we make predictions?

All we need to do is predict the response Y on the basis of a set of values for the predictors $X_1, X_2, \ldots, X_p$. However, there are three possible uncertainties associated with this prediction:

1. The coefficient estimates $\hat{\beta}$ are estimates for β. Mathematically,

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_p$$

is only an estimate for

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

because of the reducible error mentioned previously. We will compute a confidence interval in order to determine how close $\hat{Y}$ is to $f(X)$.

2. We also mentioned model bias earlier, which is very applicable to linear models. Sometimes assuming linear models can add reducible error, especially if the true relationship is nonlinear. When using a linear model, we are estimating the best linear approximation to the true surface.

3. Even if $f(X)$ was given and the true values for the coefficients β, the random, irreducible error ϵ will prevent the response value from being predicted perfectly. This is why prediction intervals are used; they incorporate both the error in the estimate for $f(X)$ and the uncertainty as to how much an individual point will differ from the population regression plane (irreducible error).

Suppose, on the *Advertising* data set, that \$100,000 is spent on TV advertising and \$20,000 is spent on radio advertising in each city.

- The 95% confidence interval is [10,985, 11,528]. That is to say, 95% of intervals of this form will contain the true value of the average sales over a large number of cities.

- The 95% prediction interval is [7,930, 14,580]. That is to say, 95% of the intervals of this form will contain the true value of sales in a given city.

Both intervals are centered at 11,256, but the prediction interval is much wider than the confidence interval, suggesting increased uncertainty about sales in comparison to the average sales over many locations.

3.2.3 Considerations in Linear Regression Models

So far, we have only dealt with quantitative predictors, but it is also very common to have qualitative predictors. In statistical modeling, yes/no variables are coded as 1's and 0's, respectively. For example, if X is a qualitative predictor on whether you are a California resident, then $X = 1$ means you are a resident (yes) and $X = 0$ means you are not a resident (no). Suppose Y is the amount of time a week (in minutes) that someone spends outdoors. The simple linear model is

$$Y = \beta_0 + \beta_1 X + \epsilon.$$

Now $X_i = \{0, 1\}$, where 1 is a yes and 0 is a no. Estimating β_0 and β_1 is the same as before (minimizing the RSS with respect to the β's). The estimated regression equation is

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X.$$

We can also think of $\hat{Y}$ as the following:

$$\hat{Y} = \begin{cases} \hat{\beta}_0 + \hat{\beta}_1 & \text{if } X = 1 \\ \hat{\beta}_0 & \text{if } X = 0 \end{cases}$$

If $\hat{\beta}_0 = 45$ and $\hat{\beta}_1 = 15$, then we would predict a California resident to spend, on average, 15 more minutes outside compared to non-California residents. The standard statistical measures still apply (p-value, t-statistic, F-statistic for multiple linear regression). If the p-value for $\hat{\beta}_1$ was greater than 0.05 at the 0.05 significance level, then we would conclude that there is no difference in outside time between California and non-California residents.

The resident regression model is an example of a qualitative predictor with $k = 2$ levels. Some qualitative variables will have more than 2. Consider the *ThreeCars* data set of the following properties:

- Contains $n = 90$ observations.
- Each vehicle has its make recorded: Porsche, Jaguar, and BMW.
- Quantitative predictors include the cars' recorded age (years) and mileage (in thousands of miles).
- In our case, we will predict the car's value (in thousands of USD).

Suppose we fit a linear regression using the vehicle models as the predictor and value as the response. Instead of fitting 2 coefficients, we will fit 3 to account for the $k = 3$ levels of the qualitative predictor. In general, for a qualitative predictor with k levels, $k - 1$ coefficients will need to be optimized. Let X_1 and X_2 represent whether the owner has a Jaguar or BMV, respectively. The appropriate linear regression model becomes

$$Y = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if the person owns a Jaguar} \\ \beta_0 + \beta_2 + \epsilon_i & \text{if the person owns a BMV} \\ \beta_0 + \epsilon_i & \text{if the person owns a Porsche} \end{cases}$$

Notice how if $X_{i1} = 0$ and $X_{i2} = 0$, we can automatically deduce that the owner's car is a Porsche. The variables X_{i1}, X_{i2} are *dummy variables*, representing the presence (0 vs. 1) of a category. The level, or category, not associated with a dummy variable is known as the *baseline* (Porsche in our case).

	Estimate	Std. error	t-statistic	p-value
Intercept	30.233	2.775	10.896	< 0.0001
CarTypeJaguar	1.723	3.924	0.439	0.662
CarTypePorsche	20.303	3.924	5.174	< 0.0001

With the qualitative predictor alone, it is estimated that the BMW's value is \$30,233. The Jaguar is estimated to have a mean value \$1,723 higher than BMW and the Porsche is estimated to have a mean value \$20,303 above the BMW. Judging by the p-values, there is no statistical evidence of a real difference in average value between a BMW and a Jaguar. The F-statistic for this model is 16.46, implying that at least one predictor is associated with car value. In this case, the significance is driven by the Porsche predictor.

Extensions of the Linear Model

The two most important assumptions imposed on a linear model is *additivity* and *linearity*.

- The *addivitity* assumption means that the association between a predictor X_i and the response Y does not depend on the values of the other predictors.
- The *linearity* assumption states that the change in the response Y associated with a one-unit change in X_i is constant, regardless of the value of X_i.

We discuss common extensions of linear models; some of which require us to break these underlying assumptions.

For example, in the *Advertising* data, we concluded that both TV and radio seem to be associated with sales. This simple model may be incorrect, however. Suppose spending money on radio advertising actually increases the effectiveness of TV advertising, so that the slope term for TV should increase as radio increases. In such a situation, given a fixed budget of \$100,00, spending half on radio and half on TV may increase sales more than allocating the entire amount to either TV or to radio. In marketing, this is a *synergy*, or *interaction* effect. Recall the standard regression model with two variables:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

According to this model, a one-unit increase in X_1 is associated with an average increase in Y of β_1 units. The presence of X_2 does not alter this statement, which is why we say is this increase occurs while holding X_2 fixed. One way of extending is model is to include a third predictor, called an *interaction term*, which is constructed by computing the product of X_1 and X_2. The new model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \epsilon.$$

If we rewrite factor out X_1 (or X_2), we get

$$Y = \beta_0 + X_1(\beta_1 + \beta_3 X_2) + \beta_2 X_2 + \epsilon.$$

The association between X_1 and Y is no longer constant; a change in the value of X_2 will change the association between X_1 and Y. The same applies to changing X_1 and the association between X_2 and Y.

If we wanted to include an interaction term with radio and sales, we get

$$Y = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{radio} + \beta_3(\text{TV} \times \text{radio})$$

	Coefficient	Std. error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

Table 3.12: Taken from *An Introduction to Statistical Learning: Applications in Python*

We interpret β_3 as the increase in effectiveness of TV advertising associated with a one-unit increase in radio advertising (or vice-versa). The results in Table 3.12 strongly imply that the model including the interaction term is superior to the model containing only the main effects. We have evidence for $H_a : \beta_3 \neq 0$ (p-value < 0.05), suggesting the true relationship is not additive. The R^2 is 0.968 compared t 0.897 with only the main effects. This means $\frac{96.8-89.7}{100-89.7} = 69\%$ of the variability in sales that remains after fitting the additive model has been explained by the interaction term.

The coefficients explain that:

- An increase in \$1,000 of TV advertising is associated with increased sales of $19.1 + (1.1 \times \text{radio})$ units.
- An increase in \$1,000 of radio advertising is associated with increased sales of $28.9 + (1.1 \times \text{TV})$ units.

This is an example in which the p-values associated with each term is statistically significant, providing sufficient evidence to keep them in the model. Sometimes the interaction term's p-value is very small but the associated main effects do not. According to the hierarchial principle states that if we include an interaction in the model, we should also include the main effects, even if the p-values associated with their coefficients are not significant. That is to say, if the interaction between X_1 and X_2 seems important, then we should include both X_1 and X_2 even if their coefficient estimates have large p-values. If $X_1 \times X_2$ is related to the response, then we do not care whether the coefficients for X_1 and X_2 are very close to zero. Also, $X_1 \times X_2$ is typically correlated with X_1 and X_2, and so leaving them out tends to alter the meaning of the interaction.

Interaction terms can also feature qualitative predictors. The *Depression* data set entails the following:

- A pharmaceutical company had a study conducted of the effect of two different types of treatments on depression that they are developing.

- 36 subjects were randomized to receive treatment A or treatment B and had their age and initial depression severity recorded.
- After 3 months of treatment, each subject received an effectiveness score (how well the treatment worked in lowering depression)
- Y = effectiveness score, X_1 = age (years), and X_2 = treatment B yes/no (TRTB will be 1/yes, meaning treatment B, and 0/no means treatment A).

It can be hypothesized that the effect of the treatment is not the same on subjects. Additionally, it can be further hypothesized that the effect of the treatment differs across the different ages (this will add a $\beta_3(\text{age}_i \times \text{TRTB}_i)$ term). The resulting model to predict the effectiveness score is

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \text{age}_i + \hat{\beta}_2 \text{TRTB}_i + \hat{\beta}_3(\text{age}_i \times \text{TRTB}_i).$$

	Estimate	Std. error	t-statistic	p-value
Intercept	47.5156	4.8471	9.803	< 0.0001
age	0.3305	0.1033	3.201	0.00309
TRTB	-31.5774	5.8051	-5.440	< 0.0001
age:TRTB	0.4842	0.1243	3.895	0.00047

Table 3.13

- For the treatment A group, $\hat{Y}_i = 47.51 + 0.33X_{1i}$.
- For the treatment B group, $\hat{Y}_i = 15.94 + 0.81X_{1i}$.
- The p-value associated with the hypothesis test $\beta_3 = 0$ is 0.00047; thus, we reject the null and conclude the alternative, that $\beta_3 \neq 0$. This implies that the effect on age on the response (effectiveness score) depends on the treatment or vice versa, that the effect of the treatment type (A or B) on the response depends on the subject's age.

Non-linear Relationships

In some cases, the true relationship between the response and the predictors may be non-linear. A common extension to combat the linearity assumption is using polynomial regression, or a polynomial transformation of X (ex. X^2, X^3, X^4, and so on).

We will look at the *SAT* data set as an example. The response variable Y is the average SAT score, with potential associated predictors X_1, X_2 as the % taking the exam out of all eligible (takers) and the amount spent by the state for public secondary schools, per student ($100's), respectively. What if we fitted the simple linear regression model

$$Y = \beta_0 + \beta_1 \text{Takers} + \epsilon?$$

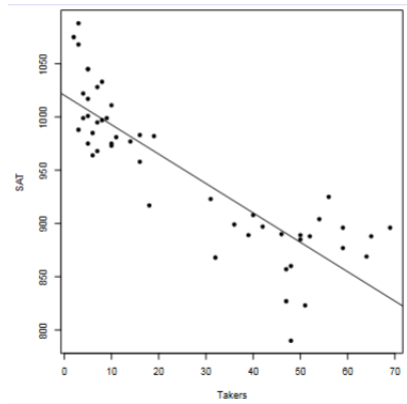


Figure 3.14

Judging by Figure 3.14, there is potentially a curved relationship between X and Y. Taking this consideration into account, we now fit a *full second-order model*

$$Y = \beta_0 + \underbrace{\beta_1 X_1 + \beta_2 X_2}_{\text{First order}} + \underbrace{\beta_3 X_1^2 + \beta_4 X_2^2}_{\text{Second order}} + \underbrace{\beta_5 X_1 X_2}_{\text{Interaction}} + \epsilon$$

	Estimate	Std. error	t-statistic	p-value
Intercept	893.66283	36.14094	24.727	< 0.0001
takers	-7.05561	0.83740	-8.426	< 0.0001
takers^2	0.07725	0.01328	5.816	< 0.0001
expend	10.33333	2.49600	4.140	0.000155
expend^2	-0.11775	0.04426	-2.660	0.010851
takers:expend	-0.03344	0.03716	-0.900	0.373087

Table 3.15

The R^2 for the simple linear regression model is 0.7358 and 0.8997 for the full second-order model, implying that a model that uses these additional terms to predict the average SAT term score is substantially better than one that only uses the percentage of eligible takers. If we tested for $H_0 : \beta_5 = 0$, we obtain the p-value of $0.373087 > 0.005$. We fail to reject the null and conclude no evidence that the interaction term needs to be in the model.

Note that adding more terms, or degrees to the model may not result in a better fit to the data. A fifth-order model $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \beta_5 X^5$ can be more "wiggly" than a second-order model and perform worse on test data due to its higher flexibility.

Potential Problems for Linear Models

Fitting a linear regression model to a particular data set can lend itself to a variety of problems. We discuss each of the following in detail:

1. Non-linearity of the response-predictor relationships.
2. Correlation of error terms.
3. Non-constant variance of error terms.
4. Outliers.
5. High-leverage points.
6. Collinearity.

1. Non-linearity of the Data

The linear regression model assumes a straight-line relationship between the predictors and the response. If the true relationship is far from linear, then all of our conclusions will be suspicious!

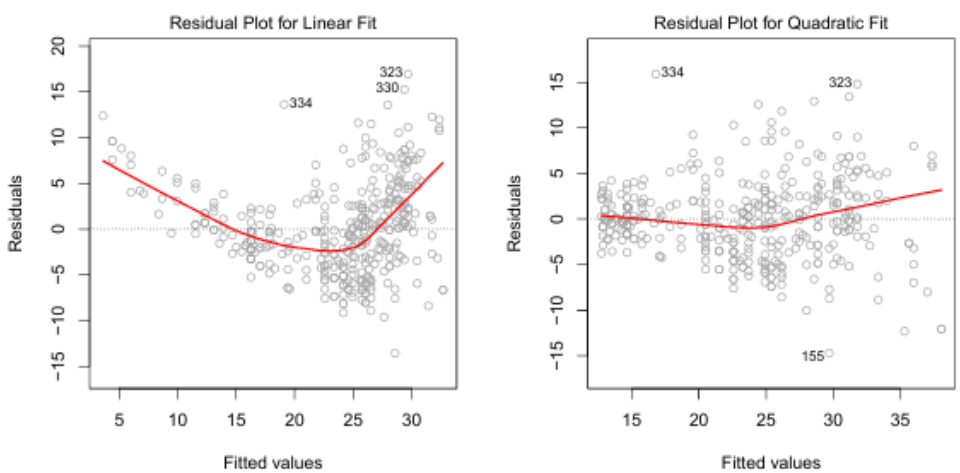


Figure 3.16: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 3.16 displays a *residual plot*, serving as a useful graphical tool for identifying non-linearity. We can plot residuals $y_i - \hat{y}_i$ versus the predictor x_i. In this case of a multiple regression model, there are multiple predictors, we instead plot the residuals versus the predicted (or *fitted*) values $\hat{y}_i$. The presence of a pattern may indicate a problem with some aspect of the linear model, but ideally the residual plot will show no discernible pattern.

The left-hand panel of Figure 3.16 shows a residual plot for a linear model fitted to a certain data set. The residuals clearly exhibit a U-shape, providing a strong indication of strong non-linearity in the data. On the other hand, the right-hand panel displays the residual plot for a quadratic model fitted to a certain data set. There is no definite pattern in the residuals, suggesting that the quadratic model is a better fit for the data. Typically, some type of transformation (ex. $\log X$, $\sqrt{X}$, X^2) is imposed if we observe patterns in the residual plots.

2. Correlation of Error Terms

A critical assumption of linear regression models is that the error terms $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$, are uncorrelated. That is to say, the value of ϵ_i provides little to no information about the value of ϵ_{i+1}. The standard errors computed for the estimated regression coefficients (or fitted values) are based on the assumption of uncorrelated error terms. Should correlation exist, the estimated standard errors will generally underestimate the true standard errors, lending confidence and prediction intervals to be tighter. Therefore, if the error terms are correlation, we may have an unwarranted sense of confidence in our model.

For example, if we doubled our data, leading to observations and error terms identical in pairs, our SE calculations would be as if we had a sample size of $2n$, narrowing the confidence intervals by a factor of $\sqrt{2}$.

This concern is very prevalent in *time series analysis* (to be discussed later), which consists of observations for which measurements are obtained at discrete points in time. In many cases, observations that are obtained at adjacent time points will have positively correlated errors.

3. Non-constant Variance of Error Terms

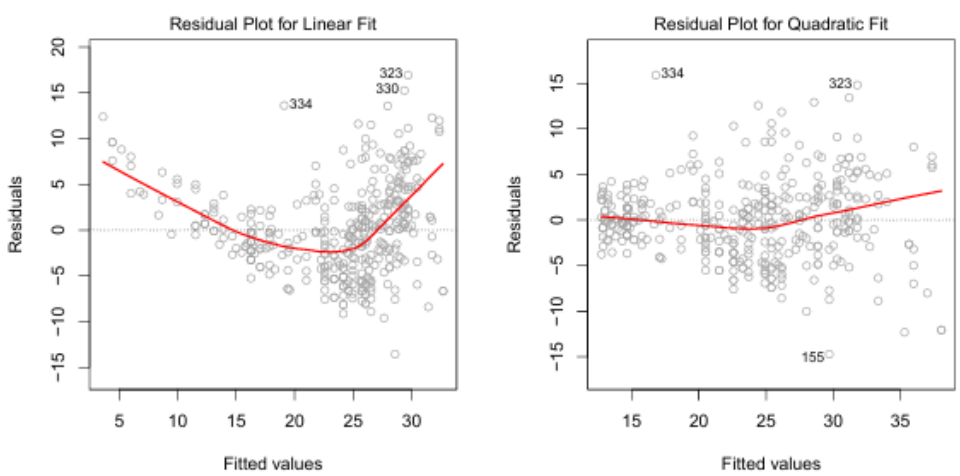


Figure 3.17: Taken from *An Introduction to Statistical Learning: Applications in Python*

We also assumed the error term have a constant variance, or $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. The standard errors, confidence intervals, and hypothesis tests associated with linear models rely upon this assumption.

Unfortunately, this is not usually the case. As an example, the variances of the error terms may increase with the value of the response. The left-hand panel of Figure 3.17 exhibits this pattern, known as **heteroscedasticity**, displaying a *funnel shape* in the residual plot. When faced with such a problem, one possible solution is to transform the response Y using a concave function such as $\log Y$ or $\sqrt{Y}$, resulting in a greater amount of shrinkage in the larger responses and a reduction in heteroscedasticity.

In other cases, we may have a good idea of the variance of each response. For example, if the ith response could be an average of n_i raw observations. If each raw observation is uncorrelated with with variance σ^2, then their average has variance $\sigma_i^2 = \frac{\sigma^2}{n_i}$. In this case a simple remedy is fit our model by *weighted least squares*, with weights proportional to the inverse variances ($w_i = n_i$).

4. Outliers

An **outlier** is a point for which y_i is far from the value predicted by the model. Outliers can arise for many reasons, such as incorrect recording of an observation during data collection.

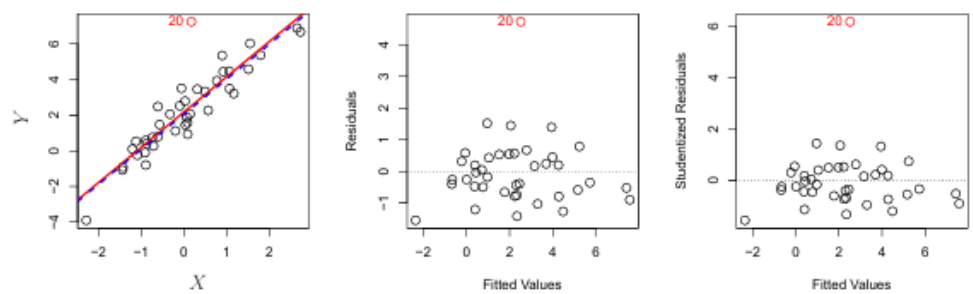


Figure 3.18: Taken from *An Introduction to Statistical Learning: Applications in Python*

The red point (obs. 20) in the left-hand panel of Figure 3.18 illustrates a common outlier. In this case, removing the outlier has little effect on the least squares line. While this is common for most outliers, other statistical measurements are greatly affected. The RSE can drop significantly when the outlier is removed, narrowing confidence intervals, decreasing the associated p-values, and increasing the R^2.

Residual plots are also used to identify outliers (see middle panel of Figure 3.18). In practice, it can be difficult to decide how large a residual needs to be before we can consider the point to be an outlier. We can instead plot the studentized residuals, obtained by dividing each residual e_i by its estimated standard error. Generally, we regard observations with its studentized residual between -3 and 3 an outlier. As

observed in the right-hand panel of Figure 3.18, obs. 20 has a studentized error of roughly 6 while all of the other observations lie between -2 and 2.

Outliers believed to have been caused by errors in data entry, collection, or recording can be solved by removing them. One should proceed with caution; outliers may be attributed to deficiencies in the model (ex. a missing predictor).

5. High Leverage Points

As outliers have an extreme value in the response variable, observations with **high leverage** have an extreme value in x_i.

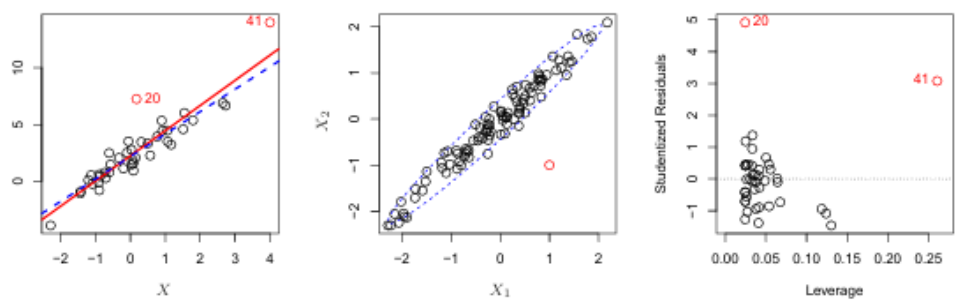


Figure 3.19: Taken from *An Introduction to Statistical Learning: Applications in Python*

As seen in the left-hand panel of Figure 3.19, obs. 20 is not a leverage point because its value in X is common among the other observations (it could be in contention as an outlier, however). Obs. 41 would be regarded as a high leverage point because its value in X is large relative to the other observations. The dashed line is the least squares fit when obs. 41 is removed, resulting in a much more substantial impact on the least squares line than removing the outlier. In general, high leverage observations have a sizable impact on the estimated regression line and a greater cause for concern if the least squares line is heavily affect by just a couple of observations.

High leverage points are very easy to identify in simple linear regression. However, in multiple linear regression, it is possible to have an observation that is well within the range of each individual predictor's values, but unusual in terms of the full set of predictors. The middle panel of 3.19 show a data set with two predictors X_1, X_2. Most of the observations' predictor values fall within the dashed ellipse, but the red observation is well outside of this range. By examining X_1 or X_2 individually, we will not notice this high leverage point. This problem is typically more pronounced in multiple linear regression, with no way to plot all dimensions simultaneously.

Fortunately, we are able to quantify an observation's leverage. The value

$$\boxed{h_i = \frac{1}{n} + \frac{(x_i - \overline{x})^2}{\sum_{i'=1}^{n}(x_{i'} - \overline{x})^2}}$$

is the *leverage statistic*; a large value of this statistic indicates an observation with high leverage. It is clear that h_i increases with the distance of x_i from $\overline{x}$. The following items concerning the leverage statistic hold:

1. $$\frac{1}{n} \leq h_i \leq 1$$

2. The average leverage for all the observations is always equal to $\frac{p+1}{n}$.

3. If a given observation has a leverage statistic greatly exceeding $\frac{p+1}{n}$, we may suspect that the corresponding point has high leverage.

The right-hand panel of Figure 3.19 reveals a dangerous combination: obs. 41 has both a high studentized residual as well as high leverage! Also, we see why obs. 20 has minimal impact on the least squares line: it has a large studentized residual but almost no leverage.

6. Collinearity

Collinearity refers to the situation in which two or more predictor variables are closely related to one another. There are two types of collinearity:

- *Structural collinearity*: A mathematical artifact caused by creating new predictors, such as, creating the predictor X^2 from the predictor X or creating interactions.
- *Data-based collinearity*: The result of a poorly designed experiment, reliance on purely observational data, or the inability to manipulate the system on which the data are collected.

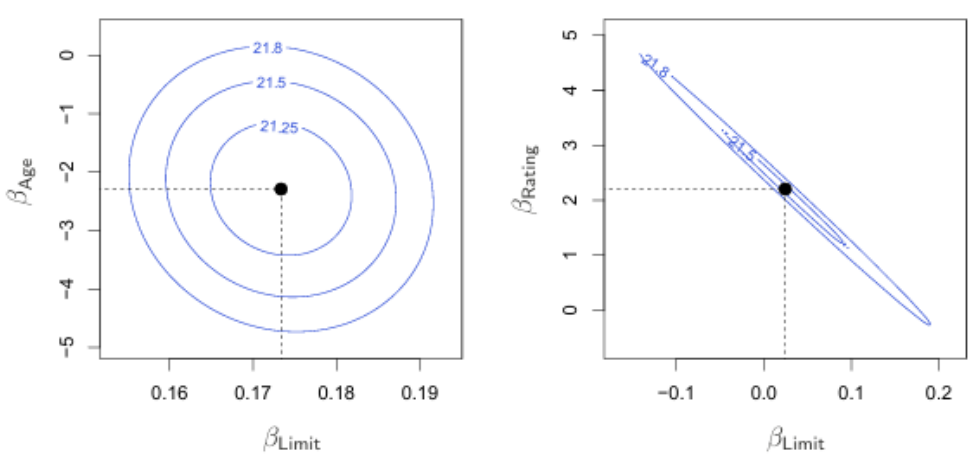


Figure 3.20: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 3.20 illustrates some of the difficulties that can result from collinearity. The left-hand panel is a contour plot of the RSS associated with different possible coefficient estimates for the regression of the response on two predictors. Each ellipse represents a set of coefficients that correspond to the same RSS, with ellipses nearest to the center taking on the minimal values of the RSS. The black dots are the "coordinate positions" of the coefficients with the minimized RSS. Thus the plot includes all plausible values for coefficients.

If features are collinear, the contours run along a narrow valley as observed in the right-hand panel of Figure 3.20. There is a broad range of values for the coefficient estimates that result in equal values for RSS. Hence a small change in the data could cause the pair of coefficient values that yield that smallest RSS—that is, the least squares estimate—to move anywhere along this valley.

Such collinearity reduces the accuracy of the estimates of the regression coefficients; it causes the standard error for $\hat{\beta}_j$ to inflate. Because the t-statistic is inversely proportional to the standard error, we expect the t-statistic to decline. As a result, in the presence of collinearity, we may fail to reject $H_0 : \beta_j = 0$. This means that the probability of detecting a *non-zero* coefficient is reduced by collinearity.

One approach is to look at the correlation matrix of the predictors. An element of this matrix that is large in absolute value indicates a pair of highly correlated variables, and therefore a collinearity problem in the data. Unfortunately, not all collinearity problems can be detected by inspection of the correlation matrix, especially if it exists between three or more predictors (called **multicollinearity**). Assessing multicollinearity can be computed using the variance inflation factor (VIF). The VIF is used to determine if a predictor is highly correlated with the other predictors. For the p-th predictor, the VIF is

$$\boxed{\text{VIF}_p = \frac{1}{1 - R_p^2.}}$$

where R_p^2 is the R-squared value obtained by regressing the p-th predictor on the remaining predictors. In other words, VIF can be defined in multiple ways:

- How much the variance of $\hat{\beta}_p$ is inflated when it (the p-th predictor) goes from being the only predictor in the model to being one of the many in the full model.
- How much larger the variance increase compared to if that variable had 0 correlation to other predictors in the model.
- The ratio of the minimum the variance of $\hat{\beta}_p$ (from just a model using the single predictor) can be and the full variance when the other predictors are in the model as well.

The general rule of thumb is that we are looking to see if VIF > 10. This would suggest that the predictor is highly related to the others.

There are two simple solutions when facing collinearity.

1. Drop one of the problematic variables from the regression. This can usually be done without much compromise to the regression fit, since the presence of collinearity implies the information that this variable provides about the response is redundant in the presence of other variables.

2. Combine the collinear variables together into a single predictor.

3.3 The Marketing Scenario

We return to the questions posed at the start of this chapter:

1. *Is there a relationship between advertising and budget sales?* This can be answered by fitting a multiple regression model and testing the null hypothesis $H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$.

2. *How strong is the relationship between advertising budget and sales?* The two measures of model accuracy used in linear regression models are the RSE and R^2 statistic.

3. *Which media are associated with sales?* To answer this question, we can examine the p-values associated with each predictor's t-statistic.

4. *How large is the association between medium and sales?* We saw that the standard error of $\hat{\beta}_j$ can be used to construct confidence intervals for β_j. Moreover, standard errors can get extremely wide due to collinearity between two or more predictors, measurable through VIFs.

5. *How accurately can we predict future sales?* The accuracy associated with this estimate depends on whether we wish to predict an individual response, $Y = f(X) + \epsilon$, or the average response, $f(X)$. If the former, use a prediction interval, and if the latter, we use a confidence interval. Prediction intervals will always be wider than confidence intervals because they account for the uncertainty associated with ϵ, the irreducible error.

6. *Is the relationship linear?* Residual plots can be used to identify non-linearity. If a pattern exists, transformations can be applied to make the residual line more "linear."

7. *Is there synergy among the advertising media?* While the standard linear regression model assumes additivity between the predictors the response, it can be unrealistic for particular data sets. Adding interaction terms are used to accommodate for the

effect that two predictors can have on each other; a small p-value associated with the interaction term indicates the presence of these non-additive relationships.

3.4 Comparison with K-Nearest Neighbors

After witnessing two models—K-nearest neighbors and linear regression, we will compare and two and discuss their advantages over each other.

There are a few underlying distinctions between the two models:

- K-nearest neighbors is a *non-parametric* method and linear regression models are parametric. Because linear regression assumes a functional form for $f(X)$, it is less flexible than K-nearest neighbors.
- K-nearest neighbors has a hyperparameter (K: the number of neighbors) that can be adjusted to improve the model's fit and accuracy. Recall that when $K = 1$, the model will perfectly fit the training data, but perform worse on test data from overfitting. Plain linear regression models do not have hyperparameters; any hyperparameters will be from resampling or through an iterative method to optimize coefficients.
- Linear regression is easier to interpret, primarily because of the functional form it assumes. We can easily predict the response by plugging in the predictor values into $\hat{f}(X)$. K-nearest neighbors lacks a clear model structure, making it difficult to understand the value of a prediction without knowing the k closest observations.

The big question is: in what setting will a parametric approach such as least squares linear regression outperform a non-parametric approach such as KNN regression? The simple solution is:

The parametric approach will outperform the non-parametric approach if the parametric form that has been selected is close to the true form of f.

For a model whose relationship is truly linear, then linear regression will almost always outperform KNN, regardless of the number of neighbors chosen. As we transition to more non-linear relationships, KNN will begin to surface over linear regression in fitting the exact relationship.

The left-hand panel of Figure 3.21 shows the KNN fit with $K = 1$ (blue) and $K = 9$ (red) alongside the true relationship for a data set with $n = 50$ observations. The right-hand panel shows the mean squared error of both linear regression (dashed line) and the KNN fit from $K = 1$ to $K = 9$. We observe that for $K \geq 4$, KNN outperforms linear regression. When the true relationship is very non-linear as shown in Figure 3.22, there is a significant drop in the MSE when we shift from using a linear regression model to a KNN regression model to fit the data. Notice that the test MSE remains relatively unchanged

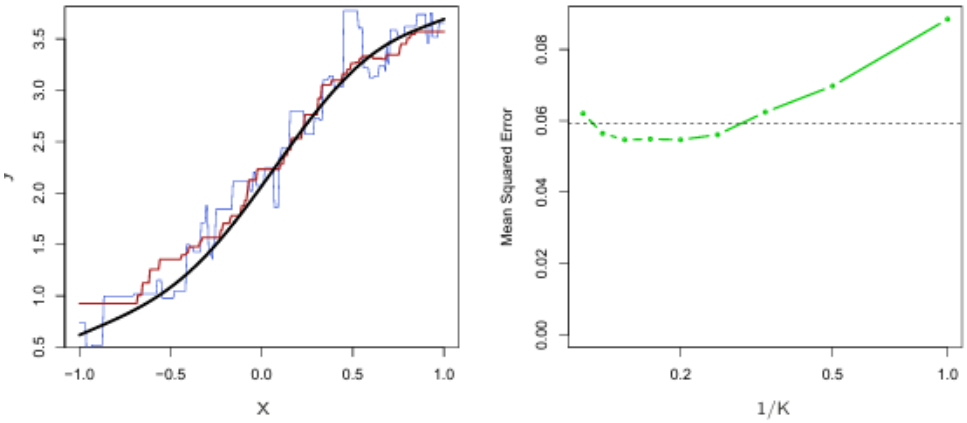


Figure 3.21: Taken from *An Introduction to Statistical Learning: Applications in Python*

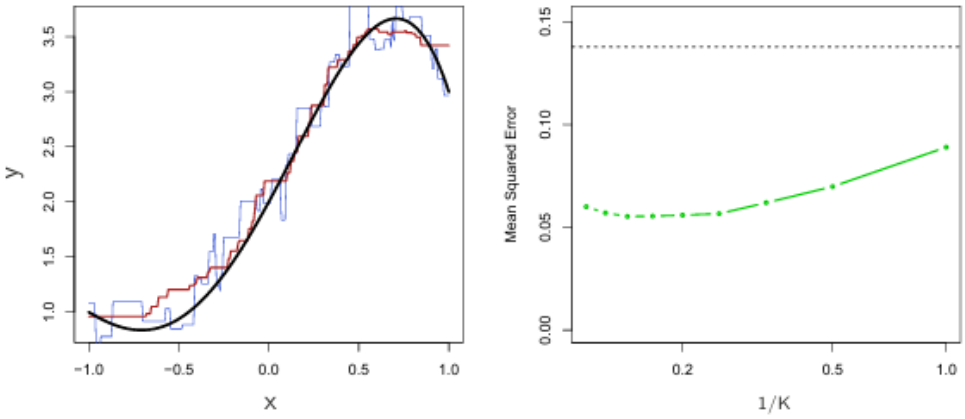


Figure 3.22: Taken from *An Introduction to Statistical Learning: Applications in Python*

between both figures, but the linear regression MSE grows substantially.

These results do not imply, however, that KNN should always be used for extremely non-linear data. Suppose we increased the number of predictors to $p = 10$ or $p = 20$. We can reasonably expect both models to perform worse, but KNN will suffer greatly. Spreading 50 observations over $p = 20$ dimensions results in a phenomenon in which a given observation has no "nearby neighbors"– this is the so-called curse of dimensionality. That is to say, the nearest observations are very far away from x_0 when p is large, leading to a very poor prediction of $f(x_0)$ and hence a poor KNN fit. As a general rule, parametric methods will tend to outperform non-parametric approaches when the ratio $\frac{n}{p}$ is small.

Even if p is small, we may still prefer linear regression from an interpretability standpoint.

3.5 Examples

Example 3.9 (ISLP 3.7.3). Suppose we have a data set with five predictors, X_1 = GPA, X_2 = IQ, X_3 = Level (1 for College and 0 for High School), X_4 = Interaction between GPA and IQ, and X_5 = Interaction between GPA and Level. The response is starting salary after graduation (in thousands of dollars). Suppose we use least squares to fit the model, and get $\hat{\beta}_0 = 50$, $\hat{\beta}_1 = 20$, $\hat{\beta}_2 = 0.07$, $\hat{\beta}_3 = 35$, $\hat{\beta}_4 = 0.01$, $\hat{\beta}_5 = -10$.

(a) Which answer is correct, and why?

- For a fixed value of IQ and GPA, high school graduates earn more, on average, than college graduates.
- For a fixed value of IQ and GPA, college graduates earn more, on average, than high school graduates.
- For a fixed value of IQ and GPA, high school graduates earn more, on average, than college graduates provided that the GPA is high enough.
- For a fixed value of IQ and GPA, college graduates earn more, on average, than high school graduates provided that the GPA is high enough.

Solution: The third statement is correct. If the student's GPA is 3.5, then the interaction term will cancel with $\hat{\beta}_3$. If the student's GPA is greater than 3.5, then $\hat{\beta}_5 X_1 X_3 - \hat{\beta}_3 < 0$, therefore predicting a higher salary for high school students than college students.

(b) Predict the salary of a college graduate with IQ of 110 and a GPA of 4.0.

Solution:

$$\hat{Y} = 50 + 20(4) + 0.07(110) + 35(1) + 0.01(4)(110) - 10(1)(4) = 137.1$$

We expect the college graduate to earn a starting salary of \$137,100.

(c) True or False: Since the coefficient for the GPA/IQ interaction term is very small, there is very little evidence of an interaction effect. Justify your answer.

Solution: False. The magnitude of the interaction term coefficient does not matter in defining its association with the response. A more reliable indicator would be checking the coefficient's p-value.

Example 3.10 (ISLP 3.7.4). I collect a set of data ($n = 100$ observations) containing a single predictor and a quantitative response. I then fit a linear regression model to the data, as well as a separate cubic regression, i.e. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$.

(a) Suppose that the true relationship between X and Y is linear i.e. $Y = \beta_0 + \beta_1 X + \varepsilon$.

Consider the training residual sum (RSS) for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

Solution: The cubic model would be better. If the true relationship is linear then the cubic regression will have β_2, β_3 close to 0 during training, making it the same as the linear model. However, since ϵ exists, the cubic model will also have the ability to fit the ϵ better than linear.

(b) Answer (a) using test rather than training RSS.

Solution: Linear would be better because the cubic regression will try to fit the error ϵ, and if ϵ is large then the cubic model will fit the error rather than the underlying linear relationship.

(c) Suppose that the true relationship between X and Y is not linear, but we don't know how far it is from linear. Consider the training RSS for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

Solution: Because the true relationship is now non-linear a cubic regression model is more likely to fit the data compared to a linear model due to its larger flexibility (for the same reasons provided in (a)).

(d) Answer (c) using test rather than training RSS.

Solution: This is depending on the scale of non-linearity. Cubic regression would be better if the effect of non-linearity is large, otherwise linear is better.

Example 3.11. Suppose a simple linear model is fit to predict Y = weight in kilograms using X = height in centimeters of an adult. Say a new simple linear model is fit using Y = height in centimeters and X = weight in kilograms (that is to say Y and X have swapped). State whether each of the following would be the same for this new model as it was for the original model, or it would be different and explain in a sentence or two.

1. The value of R.

   Solution: Same, because the correlation because X and Y is the same as the correlation between Y and X.

2. The value of R^2.

   Solution: Same. Since R is the same, R^2 will be the same.

3. The estimate of β_1.

Solution: Different. β_1 was the expected increase in the response given a unit increase in the explanatory variable. As Y and X are changed, β_1 will differ.

4. The estimate of β_0.

   Solution: Different. Same explanation as part (c).

5. The test statistic to test if the correlation between the predictor and response is equal to 0.

   Solution: Same. The test statistic depends only on R. Since R is the same, the t-statistic will be the same.

6. The residuals for each observation.

   Solution: Different, based on the estimates of the β's being different.

Example 3.12. Assume you have data produced by a model $y = mx + \epsilon$, where ϵ is a random error term such that the mean of ϵ is 0 and the variance of ϵ is large in relation to the coefficient m. Would you expect a K-nearest neighbor model with a smaller value of k or a larger value of k to perform better?

Solution: Since the random error ϵ is large with respect to m, we do not want the solution to depend on the training data too much, thus making it better to choose a larger k.

Example 3.13 (Multiple Choice). In a linear regression analysis with the usual assumptions, which quantity is the same for each observation?

(a) The mean value of the response μ_y.

(b) The variance of the error σ_ϵ.

(c) The residual e_i.

(d) The predicted response value $\hat{Y}_i$.

Solution: B. An underlying assumption of linear models is constant variance of the error terms.

Example 3.14. Say we have a response Y which is calculated through a single quantitative predictor X. Does a very low value for R^2, say 0.01, necessarily indicate a weak relationship between X and Y?

Solution: Not quite. A low R^2 implies that the model has a lot of unexplained variance, which is generally a product of poor model choices or high variability in the data.

Example 3.15 (SOA Practice Problem 11). You are given the following results from a regression model.

Observation number (i)	y_i	$\hat{f}(x_i)$
1	2	4
2	5	3
3	6	9
4	8	3
5	4	6

Calculate the sum of squared errors (SSE).

Solution: Sum the squared difference between each response and prediction:

$$\text{SSE} = (4-2)^2 + (3-5)^2 + (9-6)^2 + (3-8)^2 + (6-4)^2 = \boxed{46.}$$

Example 3.16 (SOA Practice Problem 13). Determine which of the following statements is/are true for a simple linear relationship, $y = \beta_0 + \beta_1 x + \epsilon$.

- If $\epsilon = 0$, the 95% confidence interval is equal to the 95% prediction interval.
- The prediction interval is always at least as wide as the confidence interval.
- The prediction interval quantifies the possible range for $E[y|x]$.

Solution: The first two statements are true. For statement 1, the prediction interval includes the irreducible error, but in this case it is zero. For statement 2, the prediction interval always contains the confidence interval due to the irreducible error. Statement 3 is false; it is the confidence interval that quantifies $E[y|x]$.

Example 3.17 (SOA Practice Problem 18). For a simple linear regression model the sum of squares of the residuals is $\sum_{i=1}^{25} e_i^2 = 230$ and the R^2 statistic is 0.64.

Calculate the total sum of squares (TSS) for this model.

Solution: Use the formula for the R^2 statistic using both the RSS and TSS:

$$R^2 = 1 - \frac{RSS}{TSS}$$

$$0.64 = 1 - \frac{230}{TSS} \quad \Longleftrightarrow \quad \boxed{TSS = 638.89.}$$

Example 3.18 (SOA Practice Problem 24). Sarah performs a regression on the return on a mutual fund (y) on four predictors plus an intercept. She uses monthly returns over 105 months. Her software calculates the F-statistic for the regression as $F = 20.0$, but then it quits working before it calculates the value of R^2. While she waits on hold with the help

desk, she tries to calculate R^2 from the F-statistic.

Determine which of the following statements about the attempted calculation is true.

(a) There is insufficient information, but it could be calculated if she had the value of the residual sum of squares (RSS).

(b) There is insufficient information, but it could be calculated if she had the value of the total sum of squares (TSS) and RSS.

(c) $R^2 = 0.44$.

(d) $R^2 = 0.56$.

(e) $R^2 = 0.80$.

Solution: C. Rewrite the F-statistic with respect to the ratio $\frac{RSS}{TSS}$ (this is used in computing the R^2 statistic).

R^2-statistic	F-statistic
$R^2 = 1 - \frac{RSS}{TSS}$	$F = \frac{\frac{TSS-RSS}{p}}{\frac{RSS}{n-p-1}}$
$R^2 = 1 - \frac{RSS}{TSS}$	$20 = \frac{\frac{TSS-RSS}{4}}{\frac{RSS}{105-4-1}}$
	$20 = 25\left(\frac{TSS-RSS}{RSS}\right)$
	$0.8 = \frac{TSS}{RSS} - 1$
	$\frac{RSS}{TSS} = (1.8)^{-1}$
$R^2 = 1 - (1.8)^{-1}$	
$\boxed{R^2 \approx 0.444}$	

Example 3.19 (SOA Practice Problem 44). Two actuaries are analyzing dental claims for a group of $n = 100$ participants. The predictor variable is sex, with 0 and 1 as possible values.

Actuary 1 uses the following regression model:

$$Y = \beta + \epsilon.$$

Actuary 2 uses the following regression model:

$$Y = \beta_0 + (\beta_1 \times \text{Sex}) + \epsilon.$$

The residual sum of squares for the regression of Actuary 2 is 250,000 and the total sum of squares is 490,000.

Calculate the F-statistic to test whether the model of Actuary 2 is a significant improvement over the model of Actuary 1.

Solution: The model from Actuary 1 is the null model and so it is not needed. Use the formula for the F-statistic:

$$F = \frac{\frac{TSS-RSS}{p}}{\frac{RSS}{n-p-1}} = \frac{\frac{490000-250000}{1}}{\frac{250000}{98}} = \boxed{94.08.}$$

Example 3.20 (SOA Practice Problem 49). Trish runs a regression on a data set of n observations. She then calculates a 95% confidence interval (t, u) on y for a given set of predictors. She also calculates a 95% prediction interval (v, w) on y for the same set of predictors.

Determine which of the following must be true.

- $\lim_{n\to\infty}(u - t) = 0$
- $\lim_{n\to\infty}(w - v) = 0$
- $w - v > u - t$

Solution: Statements 1 and 3 are true. 1 is true because a confidence interval reflects the error in estimating the expected value. With an infinite amount of data, the error goes to zero. 3 is true because the additional uncertainty in making predictions about a future value as compared to estimating its expected value leads to a wider interval. 2 is false because a prediction interval also reflects the uncertainty in the prediction observation. That uncertainty is independent of the sample size and thus the interval cannot go to zero.

Example 3.21 (SOA Practice Problem 53). Determine which of the following statements is NOT true about the equation

$$Y = \beta_0 + \beta_1 X + \epsilon$$

(a) β_0 is the expected value of Y.

(b) β_1 is the average increase in Y associated with a one-unit increase in X.

(c) The error term, ϵ, is typically assumed to be independent of X.

(d) The equation defines the population regression line.

(e) The method of least squares is commonly used to estimate the coefficients β_0 and β_1.

Solution: A. β_0 is the expected value of Y when $X = 0$.

Example 3.22 (SOA Practice Problem 71). Determine which of the following statements regarding the effect of omitting important variables from the regression model specification is NOT true.

(a) It will result in underfitting.

(b) It will increase the width of prediction intervals.

(c) The coefficient estimates will remain unbiased.

(d) The estimate of the variance, s^2, is inflated.

(e) The total sum of squares remains unchanged.

Solution: C. If the omitted variable is correlated with included predictors, the coefficient estimates will be biased.

4 Linear Regression: Model Selection and Regularization

In the last chapter, we briefly exposed methods of selecting certain predictors to keep, or a subset of predictors in linear regression models. This chapter will dive into ways of choosing models and developing new fitting procedures. We will find alternative fitting methods that can yield better prediction accuracy and model interpretability, each defined as follows:

- **Prediction Accuracy**: If $n >> p$, the least squares estimate tends to have low variance. And, provided that the true relationship between the response and the predictors is approximately linear, the least squares will have low bias. If n is not much larger than p, then there can be a lot of variability in the least squares fit. If $p > n$, then a least squares estimate is unattainable! Through *constraining* or *shrinking* the estimated coefficients, we can greatly reduce the variance at the cost of a negligible increase in bias.
- **Model interpretability**: Including irrelevant variables (ones not associated with the response) leads to unnecessary complexity in the resulting model. We can remove said variables by setting their coefficient estimates to zero and obtain a more interpretable model. While the least squares estimation will almost never produce coefficient estimates of zero, methods like automatically performing *variable selection* will help us exclude irrelevant variables from a multiple regression model.

There are three common approaches to using least squares to fit a data set:

- **Subset Selection**: This approach involves identifying a subset of the p predictors that we believe is related to the response. We then fit a model using least squares on the reduced set of variables.
- **Shrinkage (or Regularization)**: This approach involves fitting a model including all p predictors. However, the estimated coefficients are shrunken towards zero relative to the least squares estimates, effectively reducing variance. Depending on the type of shrinkage, some coefficients may be estimated to be exactly zero.
- **Dimension Reduction**: This approach involves projecting the p predictors into an M-dimensional subspace, where $M < p$. This is achieved by computing M different *linear combinations*, or *projections*, of the variables. Then these M projections are used as predictors to fit a linear regression model by least squares.

While we describe these methods as it pertains to linear models, the same concepts extend to other methods.

4.1 Subset Selection

This section covers methods for selecting subsets of predictors, including the **best subsets** and **stepwise** model selection procedures.

4.1.1 Best Subset Selection

The underlying theme of a best subset selection model is fitting a separate least squares regression for each possible combination of the p predictors. That is to say, we fit all p models containing exactly one predictor, all $\frac{p(p-1)}{2}$ models that contain exactly two predictors, and so forth. We then look at all of the resulting models, with the goal of identifying the one is the "best." Identifying the best model among the 2^p possibilities is not exactly trivial; the following algorithm defines measurements for which we can compare models.

Algorithm 1 *Best Subset Selection*

1. Let $\mathcal{M}_0$ denote the *null model*, containing no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots, p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. The "best" model is defined as having the smallest RSS, or equivalently largest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using the prediction error on a validation set, C_p (AIC), BIC, or adjusted R^2. Alternatively, one can use cross-validation.

In summary, we group each model by the number of predictors they use, picking out the one with "best fit." This reduces the selection down to $p + 1$ models; the best model will be compared using one or more of the criteria mentioned in Step 3 of Algorithm 1.

WARNING: This task must be performed with care, because the RSS of these $p + 1$ decreases monotonically, and therefore the R^2 increases monotonically, as the number of predictors included in the models increases. Therefore, if we use these statistics to select the best model (namely in Step 3), then we will always end up with a model involving all of the variables.

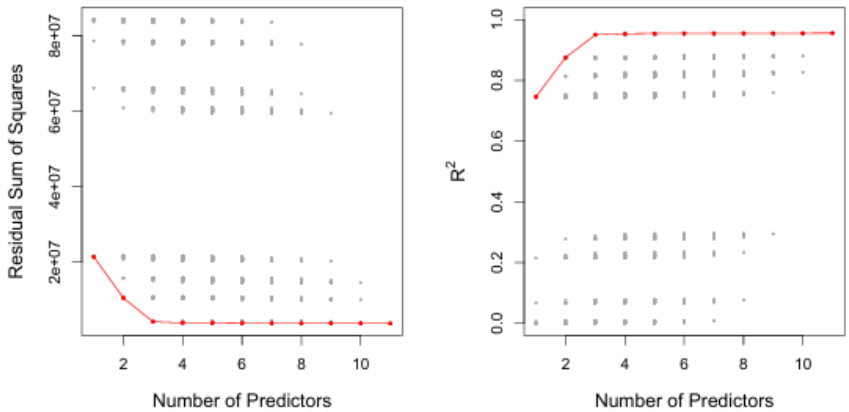


Figure 4.1: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.1 demonstrates an application of the best subsets model selection using $p = 10$ predictors, one of which is a three-level qualitative variable. Hence, there are a total of 11 possible variables which can be included. The left and right plots show the RSS and corresponding R^2 for the best models for each model size. We could conclude that, for this model, beyond $p = 3$ predictors, the RSS and R^2 do not change significantly enough to warrant more predictors.

Linear models use RSS for step 2 in the best subset selection. Other models may not use RSS as its residual measurement; said measurements would replace the RSS in step 2.

The primary drawback of applying best subsets is the computational cost associated with it. Recall that the total number of models for consideration, for p predictors, is 2^p. This quantity grows extremely fast as p increase, and becomes borderline infeasible for $p = 30$ or $p = 40$.

4.1.2 Stepwise Selection

A popular alternative to combat the computationally expensiveness of best subsets selection, or any type of subset selection, is *stepwise* selection. They explore a far more restricted set of models and thus make itself more computationally efficient.

Forward Stepwise Selection

Stepwise selection models were briefly covered in Chapter 3; we had mentioned that forward selection begins with a model containing no predictors, and then adds predictors to the model, one at a time, until all of the predictors are in the model.

Forward stepwise selection involves fitting one null model, along with $p - k$ models in the kth iteration, for $k = 0, \ldots, p - 1$. This amounts to a total of

$$\sum_{k=0}^{p-1}(p-k) = 1 + \frac{p(p+1)}{2} \text{ models.}$$

To further emphasize the efficiency, if we have a data set with $p = 20$ predictors, the best subsets selection method needs to create $2^{20} = 1{,}048{,}576$ models, whereas forward stepwise selection requires fitting only 211 models! This is because stepwise selection has quadratic dependence on p, which grows much slower in comparison to exponential growth for best subsets selection.

Algorithm 2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null model*, containing no predictors.
2. For $k = 0, 1, \ldots, p-1$:
   (a) Consider all $p-k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
   (b) Choose the "best" among these $p-k$ models, and call it $\mathcal{M}_{k+1}$. Here "best" defined as having smallest RSS or highest R^2
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using the prediction error on a validation set, C_p (AIC), BIC, or adjusted R^2. Alternatively, one can use cross-validation.

In step 2(b) of Algorithm 2, we must identify the *best* model from among those $p-k$ that augment $\mathcal{M}_k$ with one additional predictor. This can be achieved in the same way as Algorithm 6.1 (lowest RSS or highest R^2). Step 3 is more challenging; we must identify the best model among a set of models with different numbers of variables.

As mentioned previously, forward selection solves the computational intensity issue from best subset selection, but it is not guaranteed to return the best possible model out of all 2^p models. For example, suppose a data set has 3 predictors:

- The best possible model with one predictor, $\mathcal{M}_1$, contains only X_1.
- The best possible model with two predictors, $\mathcal{M}_2$, contains X_2 and X_3. In this case, forward selection will fail to select the best possible two-variable model.

Backward Stepwise Selection

Backward stepwise selection shares a common goal with forward stepwise selection: become an efficient alternative to best subset selection. However, they differ in the way their best models are obtained. The backward stepwise selection procedure begins with

the full least squares model containing all p predictors, and then iteratively removes the least useful predictor, one at a time.

Algorithm 3 *Backward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *full model*, containing all p predictors.
2. For $k = p, p-1, \ldots, 1$:
   (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k-1$ predictors.
   (b) Choose the "best" among these k models, and call it $\mathcal{M}_{k-1}$. Here "best" defined as having smallest RSS or highest R^2
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using the prediction error on a validation set, C_p (AIC), BIC, or adjusted R^2. Alternatively, one can use cross-validation.

The backward selection approach also searches through only $1 + \frac{p(p+1)}{2}$ models, and so can be applied in settings where p is too large to apply best subset selection. Additionally, this algorithm is also not guaranteed to find the best model containing a subset of the p predictors.

Keep in mind that, from the algorithm, we require $n > p$. Forward selection, on the other hand, can still operate when $n < p$ and becomes the only viable subset method when p is very large.

Hybrid Approaches

A hybrid approach incorporates both forward and backward selection. We proceed with the null model, in which variables are added to the model sequentially, in analogy to forward selection. However, after adding each new variable, the method may also remove any variables that no longer provide an improvement in the model fit.

4.1.3 Choosing the Optimal Model

Recall that the training error can be a poor estimate of the test error. For these reasons, RSS and R^2 are not suitable for selecting the best model among a collection of models with different numbers of predictors. In order to select the best model with respect to test error, we need to estimate this test error. There are two approaches:

1. We can indirectly estimate the test error by making an *adjustment* to the training error to account for the bias due to overfitting.

2. We can *directly* estimate the test error, using either a validation set approach or a cross-validation approach.

Both approaches are considered below.

C_p, AIC, BIC, and Adjusted R^2

A number of techniques for *adjusting* the training error for the model size are available and can be used to select among a set of models with different numbers of variables. The four approaches to be discussed are: C_p, *Akaike information criterion* (AIC), *Bayesian information criterion*, and *adjusted* R^2.

Definition 4.1 (Mallow's C_p). For a fitted least squares model containing d predictors, the C_p estimate of test MSE is computed using the equation

$$C_p = \frac{1}{n}(\text{RSS} + 2d\hat{\sigma}^2),$$

where $\hat{\sigma}^2$ is an estimate of the variance of the error ϵ associated with each response measurement.

Typically, $\hat{\sigma}^2$ is estimated using the full model containing all predictors. Essentially, the C_p statistic adds a penalty of $2d\hat{\sigma}^2$ to the training RSS in order to adjust for the fact that the training error tends to underestimate the test error.

The C_p statistic tends to take on a small value for models with a low test error, so when determining which of a set of models is best, we choose the model with the lowest C_p value.

Definition 4.2 (AIC/BIC criterion). The **AIC criterion** is defined for a large class of models fit by maximum likelihood. Assuming the errors are normally distributed, then

$$\text{AIC} = \frac{1}{n}(\text{RSS} + 2d\hat{\sigma}^2),$$

where the irrelevant constants have been omitted (thereby proportional to C_p). The **BIC criterion** uses

$$\text{BIC} = \frac{1}{n}(\text{RSS} + \log(n)d\hat{\sigma}^2).$$

Much like C_p, the BIC will tend to take on a small value for a model with a low test error, and so generally we select the model that has the lowest BIC value.

Note that BIC replaces $2d\hat{\sigma}^2$ used by C_p with a $\log(n)d\hat{\sigma}^2$ term, where n is the number of observations. The BIC statistic will place a heavier penalty on models with many variables and hence results in the selection of smaller models than C_p.

Also, AIC and BIC criterion are unitless. That is to say, a large AIC or BIC does not necessarily imply that the model has a poor fit on the data. Instead, these values are used in comparison only. For example, a model with an AIC of 100,000, while large, still can have a great fit to the data compared to a model with an AIC of 200,000.

Definition 4.3 (Adjusted R^2). For a least squares model with d variables, the adjusted R^2 statistic is calculated as

$$\text{Adjusted } R^2 = 1 - \frac{\frac{RSS}{n-d-1}}{\frac{SSTO}{n-1}}.$$

Recall that the usual R^2 statistic is defined as $1 - \frac{RSS}{TSS}$, where $TSS = \sum(y_i - \overline{y})^2$. Unlike the other statistics, a *large* value of adjusted R^2 indicates a model with a small test error. That is to say, maximizing the adjusted R^2 is equivalent to minimizing $\frac{RSS}{n-d-1}$. While RSS always decreases as the number of variables in the model increases, $\frac{RSS}{n-d-1}$ may increase or decrease due to the presence of d in the denominator.

The rationale behind the adjusted R^2 is that once all of the correct variables have been included in the model, adding additional *noise* variables will lead to only a very small decrease in RSS and consequently an increase in $\frac{RSS}{n-d-1}$ and a decrease in the adjusted R^2. Therefore, in practice, the model with the model with the largest adjusted R^2 will have only correct variable and no noise variables. Thus, the R^2 statistic "pays a price" for the inclusion of unnecessary variables.

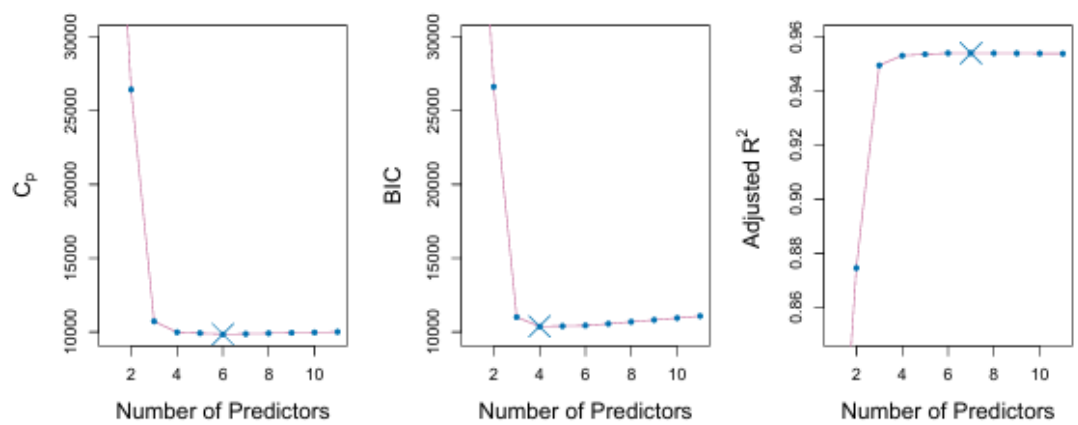


Figure 4.2: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.2 illustrates the measurements explored in Definitions 4.1-4.3 using the same data set from Figure 4.1. The marks indicate the number of predictors associated with the best fit for each given measurement. As expected, the BIC criterion suggests a model with the smallest number of predictors ($p = 4$) compared to the others.

Validation and Cross-Validation

An alternative to these approaches would be to directly estimate the test error using the training vs. test set validation and cross-validation methods discussed in Section 1.5. We can compute the error for each model (using either approach), and then select the model for which the resulting estimated test error is smallest. This has an advantage relative to the AIC, BIC, C_p, and adjusted R^2 in that it provides a direct estimate of the test error, and makes fewer assumptions about the true underlying model.

When cross-validation is used, the sequence of models $\mathcal{M}_k$ in Algorithms 1-3 is determined separately for each training fold, and the validation errors are averaged over all folds for each model sized k. Therefore, for example with best-subset regression, that $\mathcal{M}_k$, the best subset of size k, can vary across the folds. Once the best size k is picked, we find the best model of that size on the full data set.

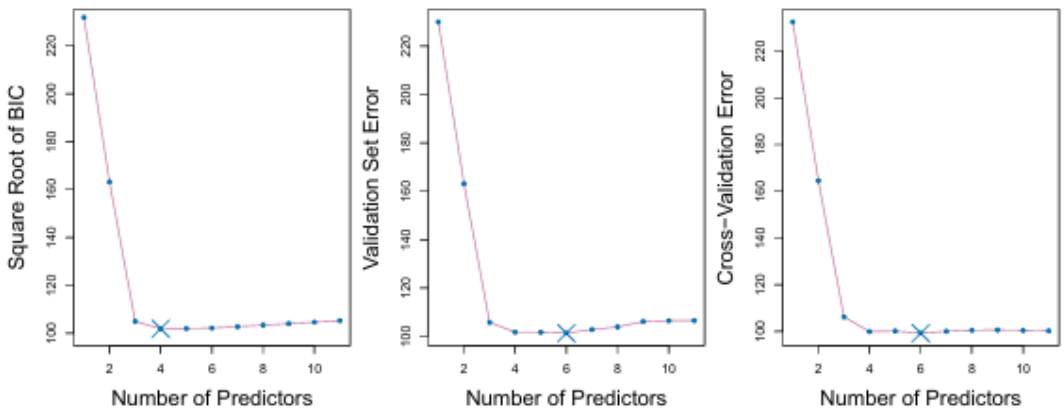


Figure 4.3: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.3 displays the BIC, test vs. training set errors, and 10-fold cross-validation errors (same data as Figure 4.1 and 4.2). The estimated test errors of the 3- to 11-variable models are quite similar for all three approaches. Both resampling methods result in a six-variable model, but all three suggest that the four-, five-, and six-variable models are roughly equivalent as it pertains to their test errors.

Should we repeat the resampling approach with different folds, then the precise model with the lowest estimated test error would surely change. In this setting, we can select a model using the one-standard error rule (1SE). We first calculate the standard error of the estimated test MSE for each model size, and then select the smallest model for which the estimated test error is within one standard error of the lowest point on the curve. The rationale here is that if a set of models appear to be more or less equally good, then we might as well choose the simplest model. Applying this methodology would lead us to select the three-variable model.

4.2 Shrinkage Methods

As an alternative to the subset selection methods is fitting a model containing all p predictors using a technique that "constrains" or "regularizes" the coefficient estimates, or equivalently, that "shrinks" the coefficient estimates towards zero. Although it may not be obvious why such constraints should improve the fit, but shrinking the coefficient estimates can significantly reduce their variance. The two best-known techniques for shrinking the regression coefficients towards zero are *ridge regression* and the *lasso*.

4.2.1 Ridge Regression

Recall that the least squares fitting procedure estimates $\beta_0, \beta_1, \ldots, \beta_p$ using the values that minimize

$$\text{RSS} = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 .$$

Ridge regression is very similar to least squares, except the coefficients are estimated by minimizing a slightly different quantity. More precisely, the ridge regression coefficient estimates $\hat{\beta}^R$ are the values that minimize

$$\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2,$$

where $\lambda \geq 0$ is a tuning parameter, to be determined separately. This equation trades off two different criteria.

1. As with least squares, ridge regression seeks coefficient estimates that fit the data well, by making the RSS small.
2. The second term is called a shrinkage penalty and is small when $\beta_1, \ldots, \beta_p$ are close to zero, and so it has the effect of "shrinking" the estimates of β_j towards zero.

λ serves to control the relative impact of these two terms on the regression coefficient estimates. When $\lambda = 0$, the penalty term has no effect, and ridge regression will produce the least squares estimates. However, as $\lambda \to \infty$, the impact of the shrinkage penalty grows, and the ridge regression coefficient estimates will approach 0.

The coefficient estimates $\hat{\beta}_\lambda^R$ will change for each value of λ; the selection of λ is very important (to be discussed later). Also, it is worth noting that the shrinkage penalty is not applied to the intercept term; we only wish to shrink the estimated association of predictors with the response. If predictors are assumed to have mean zero before ridge regression is performed, then $\hat{\beta}_0 = \overline{y}$.

The left-hand panel of Figure 4.4 displays the standardized ridge regression coefficients for a *Credit* related data set. We notice how coefficient values are varied when λ is small,

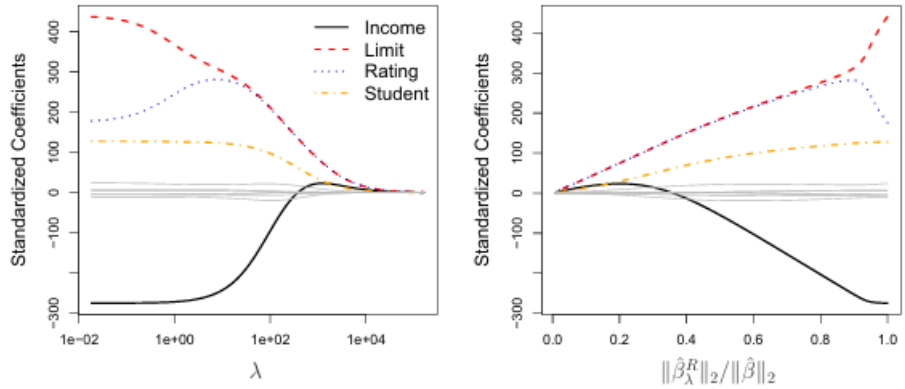


Figure 4.4: Taken from *An Introduction to Statistical Learning: Applications in Python*

but all converge to zero as λ grows larger. The x-axis of the right-hand panel is the ratio of the distances from zero for $\hat{\beta}^R_\lambda$ and $\hat{\beta}$. Therefore, since $\hat{\beta}^R_\lambda = \hat{\beta}$ when $\lambda = 0$ and $\hat{\beta}^R_\lambda \to 0$ when $\lambda \to \infty$, the domain is $(0, 1]$.

For standard least squares, multiplying X_j by a factor of k does not change the value of $X_j\hat{\beta}_j$. This does not apply to ridge regression; the value of $X_j\hat{\beta}^R_{j,\lambda}$ depends on λ and the scaling of the jth predictor. Thus, it is best to apply ridge regression after standardizing the predictors, using the formula

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}},$$

so that they are all on the same scale. The y-axis of Figure 6.4 displays the standardized ridge regression coefficient estimates—that is, the coefficient estimates that result from performing ridge regression using standardizing predictors.

Ridge regression's improvement over least squares is rooted in the *bias-variance trade-off*. As λ increases, the flexibility of the ridge regression fit decreases, leading to decreased variance but increased bias.

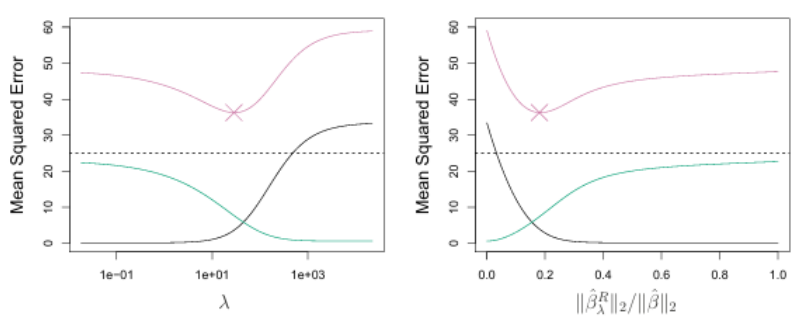


Figure 4.5: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.5 illustrates this trade-off using a simulated data set with $p = 45$ predictors and $n = 50$ observations. The green curve in the left-hand panel displays the variance of the ridge regression predictions as a function of λ. At the least squares coefficient estimates ($\lambda = 0$), the variance is high but there is no bias. However, as λ increases, the shrinkage of the ridge coefficient estimate leads to a substantial reduction in the variance of the predictions at the expense of a slight increase in bias.

- For $\lambda \in (0, 10]$, the variance decreases rapidly, dropping the MSE considerably.
- Beyond $\lambda = 10$, the decreasing in variance due to increasing λ slows, and the shrinkage on the coefficients causes them to be significantly underestimated, resulting in a large increase in the bias.

In situations where the relationship between the response and the predictors is close to linear, the least squares estimate will have low bias but may have high variance. This means a small change in the training data can cause a large change in the least squares coefficient estimates. If $p \approx n$, the least squares estimates will be extremely viable. And if $p > n$, then the least squares estimates do not even have a unique solution, whereas ridge regression can still perform well through a slight increase in bias.

Overall, ridge regression offers a substantial computational advantage compared to subset methods. It needs to only fit a single model for a fixed λ and the model-fitting procedure can be performed quickly.

4.2.2 The Lasso

The largest drawback of ridge regression is that all p predictors are guaranteed to be in the final model. While the shrinkage penalty will shift all of the coefficients towards zero, they will not be set exactly to zero (unless $\lambda = \infty$). This can cause problems in model interpretation, especially if p gets large. We may want to include a model containing only the associated predictors.

The *lasso* is a new alternative to ridge regression that overcomes this disadvantage. The lasso coefficients, $\hat{\beta}^L_\lambda$, minimize the quantity

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = \text{RSS} + \lambda\sum_{j=1}^{p}|\beta_j|$$

Ridge and lasso regression have similar formulations. The only difference is the β_j^2 in the ridge regression penalty has been replaced by $|\beta_j|$ in the lasso penalty. Lasso regression works similarly to ridge in shrinking coefficient estimates down to zero. However, the lasso's shrinkage penalty has the effect of forcing some of the coefficient estimates to be exactly zero when λ is sufficiently large. Hence, much like subset selection, the lasso

performs variable selection. This makes model interpretability much easier; as always, though, the choice of λ is crucial.

Another Formulation for Ridge Regression and the Lasso

Mathematically, the lasso and ridge regression coefficient estimates solve the respective optimization problems:

$$\underset{\beta}{\text{minimize}} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \leq s$$

and

$$\underset{\beta}{\text{minimize}} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 \leq s.$$

That is to say, for every value of λ, there is some s such that these equations will give the same lasso coefficient estimates. The same goes for the ridge coefficient estimates. For example, when $p = 2$, then this equation indicates the lasso coefficient estimates have the smallest RSS out of all points that lie within the diamond defined by $|\beta_1| + |\beta_2| \leq s$. In ridge regression, the smallest RSS lie within the circle $\beta_1^2 + \beta_2^2 \leq s$.

We can think of the constraints for the smallest RSS as a "budget" for how large $\sum_{j=1}^{p} |\beta_j|$ can be (for lasso regression). When s is very large, then this budget is not very restrictive, and so the coefficient estimates can be large. And, if s is large enough, then the equation will return the least squares solution.

A cleaner formulation that closely connects lasso, ridge, and subset regression solves the problem

$$\underset{\beta}{\text{minimize}} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^{p} I(\beta_j \neq 0) \leq s,$$

where $I(\beta_j \neq 0)$ is an indicator variable that takes on 1 if $\beta_j \neq 0$ and equals zero otherwise. This equation then turns into finding a set of coefficients such that the RSS is as small as possible, subject to the constraint that no more than s coefficients can be nonzero. However, this problem becomes practically impossible to handle once p gets large as it requires us to consider all p predictors containing s predictors.

The Variable Selection Property of the Lasso

We join the formulations with visual interpretation to understand why lasso coefficients can be forced to zero. In Figure 4.6, the least squares solution is marked by $\hat{\beta}$, while the

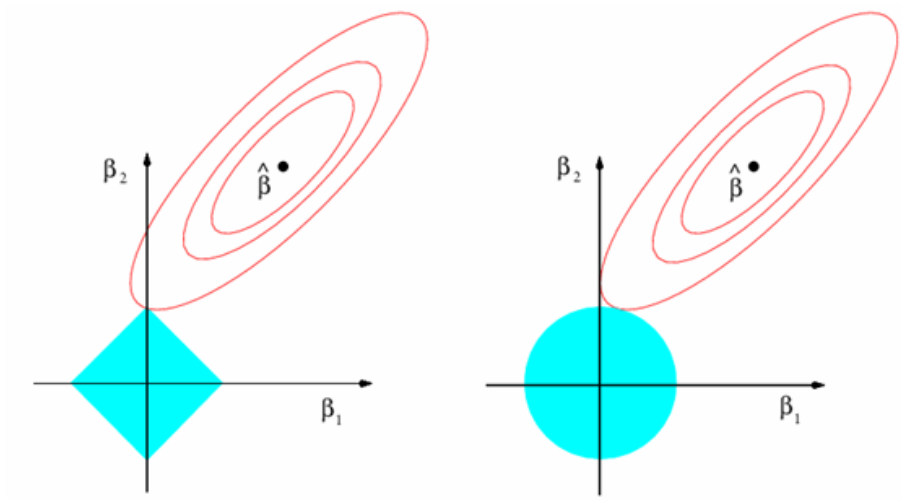


Figure 4.6: Taken from *An Introduction to Statistical Learning: Applications in Python*

blue diamond and circle represent the lasso and ridge regression constraint regions, respectively. If s is large enough, then $\hat{\beta}$ will lie inside these constraint regions, thereby making the lasso and ridge coefficient estimates equal to the least squares estimate.

Each ellipse centered around $\hat{\beta}$ are the RSS contours; all of the points on a particular ellipse have the same RSS value. As the ellipses expand away from the least squares estimates, the RSS increases. Furthermore, the optimization equations suggest that the lasso and ridge coefficients are defined by the first point at which an ellipse contacts the constraint region. The lasso constrain region has corners along each axis; when this occurs, one of the coefficients will be zero. In higher dimensions, multiple coefficients may be zero simultaneously. For ridge regression, said intersection will not typically occur on an axis, making the coefficient estimates strictly non-zero.

Lasso vs. Ridge Regression

It is certain that lasso regression has a major advantage over ridge regression in interpretation and computational cost. What about prediction and accuracy? Figure 4.7 highlights these statistics applied to the same data set as 4.5. The left-hand panel is a plot of squared bias (black), variance (green), and MSE (purple). The pattern in these statistics are very similar to ridge regression's (see Figure 4.5): as λ increases, the variance decreases and the bias increases.

On the right-hand panel of Figure 4.7, the dotted lines represent the ridge regression fits. The R^2 for both lasso and ridge are shown on the training data. The minimum MSE is slightly smaller than the of the lasso. In this data, each of the $p = 45$ predictors were related to the response, so it is no surprise that the lasso will drop some coefficients to zero and thus perform slightly worse than ridge regression.

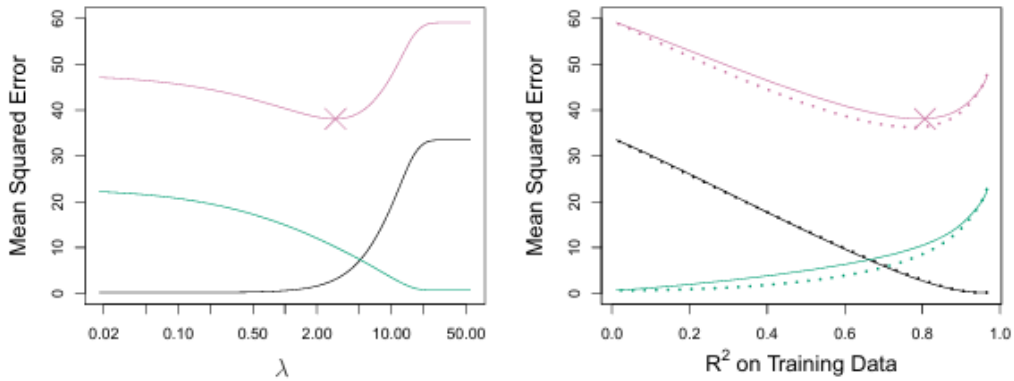


Figure 4.7: Taken from *An Introduction to Statistical Learning: Applications in Python*

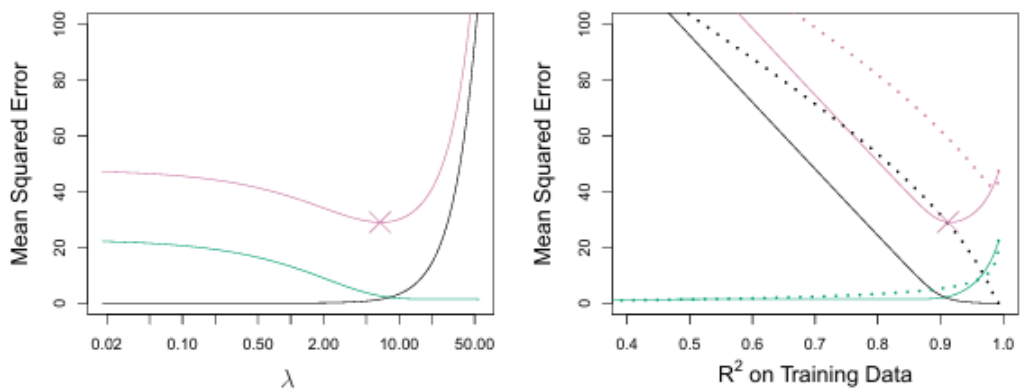


Figure 4.8: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.8 contains the same plot types as Figure 4.7, expect now only 2 of the 45 predictors are truly associated with the response. Now, the lasso is far superior to the ridge regression in terms of bias, variance, and MSE.

These examples show how one shrinkage method is not universally better than the other. We cannot identify which predictors are associated with the response. A technique such as cross-validation can be applied to determine which approach is better on a particular data set.

Both methods share a common goal of substantially lowering model variance in exchange for a small increase in bias to generate more accurate predictions. Unlike ridge regression, the lasso performs variable selection, resulting in modelings that are easier to interpret. In general, ridge regression more or less shrinks every dimension of the data by the same proportion, whereas the lasso more or less shrinks all coefficients toward zero by a similar amount, and sufficiently small coefficients are shrunken all the way to zero.

4.2.3 Selecting the Tuning Parameter

Just as subset selection requires an algorithm to determine which of the candidate models is best, implementing ridge regression and the lasso requires a method for the tuning parameter λ, or equivalently, the value of the constraint s. Cross-validation offers a simple solution to this problem. Among a grid of chosen λ values, compute the cross-validation error for each value of λ, and select the tuning parameter value for which the cross-validation error is smallest. Then, the model is re-fit using all of the available observations and the selected value of the tuning parameter.

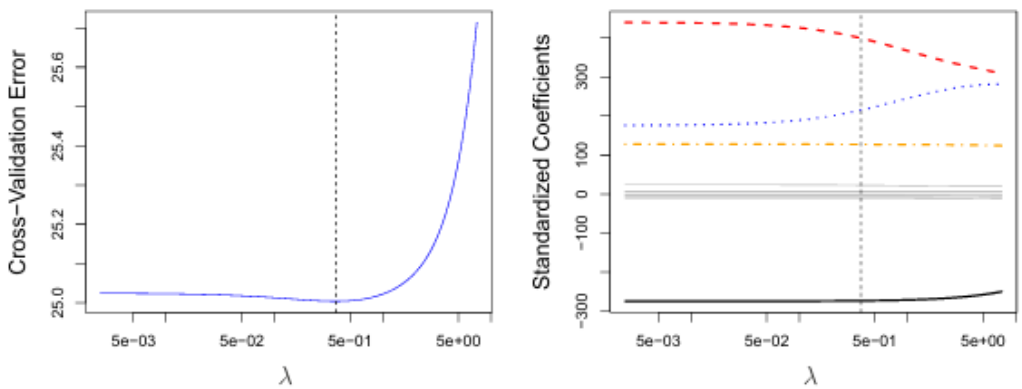


Figure 4.9: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.9 illustrates the choice of λ that results from performing LOOCV on the ridge regression fits on a given data set. The selected value of λ is given by the dashed line; in this case, the value is small, indicating that only a small amount of shrinkage is necessary relative to the least squares solution. Additionally, the dip is not very pronounced, so there is a pretty wide range of values that would give a very similar error. In this case, the least squares solution might be the best choice.

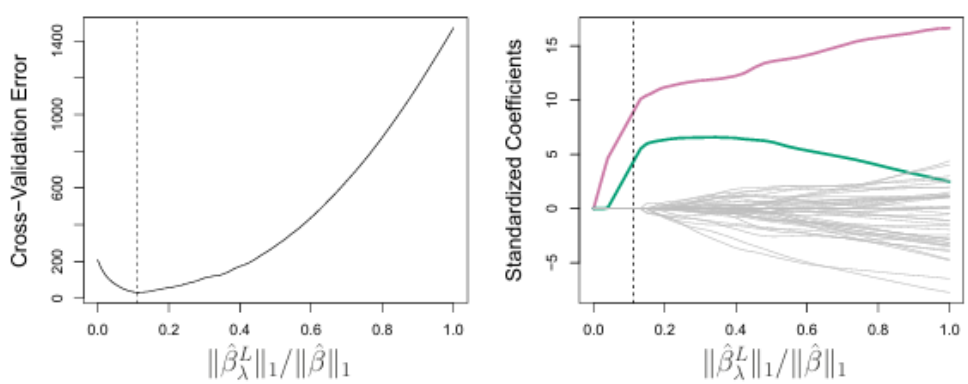


Figure 4.10: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.10 uses the same data set as Figure 4.8 (45 predictors, only 2 of which are related to the response). Using lasso regression, the dashed line indicates where the 10-fold cross-validation error is the lowest. On the right-hand panel, the lasso assigns large values to the 2 associated predictors and zero the rest at the dashed line. Even in a challenging setting, lasso regression with cross-validation successfully identified the two signal variables even with a large amount of predictors and few observations ($n = 50$).

4.3 Dimension Reduction Methods

We now explore a class of approaches that transform the predictors and then fit a least squares model using the transformed variables. These are known as *dimension reduction methods*.

Let $Z_1, Z_2, \ldots, Z_M$ represent $M < p$ linear combinations of our original p predictors. That is,

$$Z_m = \sum_{j=1}^{p} \phi_{jm} X_j$$

for some constants $\phi_{1m}, \phi_{2m}, \ldots, \phi_{pm}$, $m = 1, \ldots, M$. We can then fit the linear regression model

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \epsilon_i, \quad i = 1, \ldots, n.$$

The term *dimension reduction* stems from the fact that this approach reduces the problem of estimating the $p+1$ coefficients $\beta_0, \beta_1, \ldots, \beta_p$ to the simpler problem of estimating the $M+1$ coefficients $\theta_0, \theta_1, \ldots, \theta_M$ where $M < p$. In other words, the dimension is "mapped" from $p+1$ to $M+1$.

Note that

$$\sum_{m=1}^{M} \theta_m z_{im} = \sum_{m=1}^{M} \theta_m \sum_{j=1}^{p} \phi_{jm} x_{ij} = \sum_{j=1}^{p} \sum_{m=1}^{M} \theta_m \phi_{jm} x_{ij} = \sum_{j=1}^{p} \beta_j x_{ij},$$

where

$$\beta_j = \sum_{m=1}^{M} \theta_m \phi_{jm}.$$

Dimension reduction serves to constrain the estimated β_j coefficients as they now must take on this new form. These constraints have the potential to add bias to the coefficient estimates, but can reduce variance if $M << p$, especially in situations where p is large relative to n. If $M = p$, no dimension reduction occurs and we are essentially fitting least squares.

Dimension reductions operates in two steps:

1. The transformed predictors $Z_1, Z_2, \ldots, Z_M$ are obtained.

2. Second, the model is fit using these M predictors.

The choice of $Z_1, Z_2, \ldots, Z_M$, or equivalently, the selection of the ϕ_{jm}'s, can be achieved in different ways. The two primary approaches for this task are *principal component analysis* and *partial least squares*.

4.3.1 Principal Component Analysis

PCA is a popular approach for deriving a low-dimensional set of features from a large set of variables. While used in *unsupervised learning*, we present it here as a dimension reduction technique for regression.

Overview of PCA

PCA is a technique for reducing the dimension of an $n \times p$ data matrix $\mathbf{X}$. The first principal component direction of the data is that along with the observations vary the most.

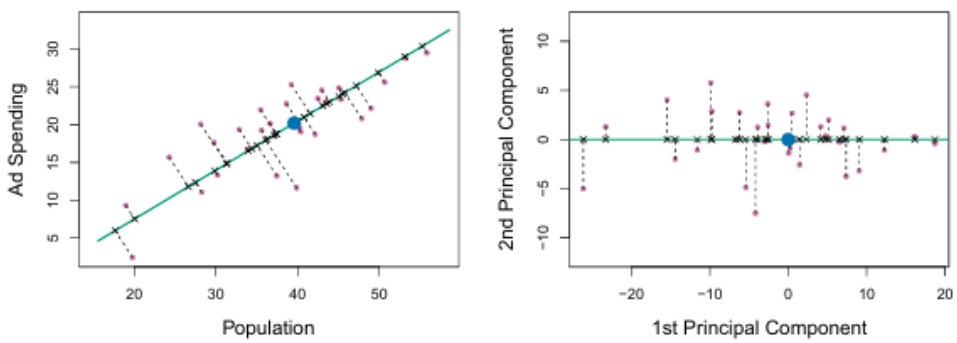


Figure 4.11: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.11 is taken from the *Advertising* data set used in Chapter 3. It shows the population size in tens of thousands of people, and ad spending for a particular company in thousands of dollars. The blue dot is the mean for pop and ad budgets. The green solid line represents the first principal component direction of the data. We can see that this is the direction along which the greatest variability in the data occurs. That is to say, we *projected* the 100 observations onto this line (left-hand panel), then the projected observations would have the largest possible variance; projecting the observations onto any other line would yield projected observations with lower variance. Projecting a point onto a line simply involves finding the location on the line which is closest to said point.

Mathematically, the reduced regression model is described by the equation

$$Z_1 = 0.839(\text{pop} - \overline{\text{pop}}) + 0.544(\text{ad} - \overline{\text{ad}}).$$

Here $\phi_{11} = 0.839$ and $\phi_{21} = 0.544$ are the principal component loadings, which define the direction referred above. $\overline{\text{pop}}$ indicates the mean of all pop values in this data set. Likewise, $\overline{\text{ad}}$ is the mean of all advertising spending. The idea is that out of every possible *linear combination* of pop and ad such that $\phi_{11}^2 + \phi_{21}^2 = 1$, this particular linear combination yields the highest variance.

The values of $z_{11}, \ldots, z_{n1}$ are known as the principal component scores, and can be seen in the right-hand panel of Figure 4.11.

Another interpretation of PCA says that the first principal component vector defines the line that is as close as possible to the data. The dashed line segments in the left-hand panel of Figure 6.15 show the sum of the squared perpendicular (or orthogonal) distances between each point and the line. Hence, the first principal component has been chosen so that the projected observations are as close as possible to the original observations.

The right-hand panel of Figure 4.11 rotates the left-hand panel so that the first principal component aligns with the x-axis. One can show that the first principal component score for the ith observation is the distance in the x-direction of the ith cross from zero. For example, the point in the bottom-left corner of Figure 4.11 has a large negative principal component score ($z_{i1} = -26.1$).

The values of the principal components Z_1 as single-number summaries of the joint pop and ad budgets for each location. In this example, if $z_{i1} = 0.839(\text{pop} - \overline{\text{pop}}) + 0.544(\text{ad} - \overline{\text{ad}}) < 0$, then this indicates a city with below average population size and below average spending; a positive score suggests the opposite. Figure 4.11 implies a linear relationship between pop and ad, so we may expect a single-number summary to work well.

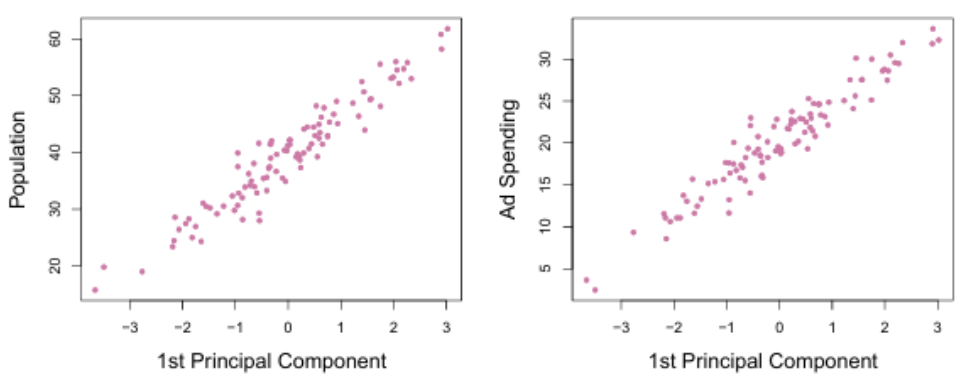


Figure 4.12: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.12 shows z_{i1} versus both pop and ad. The plots display a strong relationship between the first PC and the two features. That is to say, the first principal component appears to capture most of the information contained in the pop and ad predictors.

In general, one can construct up to p distinct principal components. The second principal component Z_2 is a linear combination of the variables that is uncorrelated with Z_1, and has largest variance subject to this constraint.

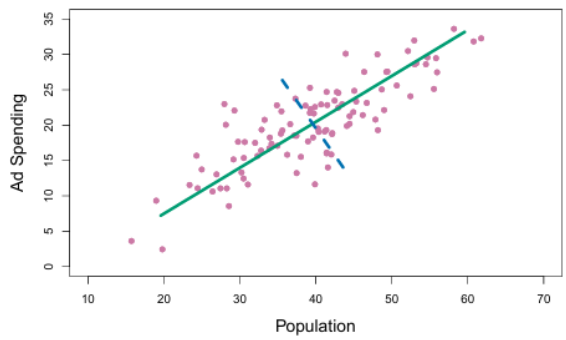


Figure 4.13: Taken from *An Introduction to Statistical Learning: Applications in Python*

In Figure 4.13, the second principal component corresponds to the blue dashed line. It turns out that the zero correlation condition of Z_1 with Z_2 is equivalent to the condition that the direction must be orthogonal to the first principal component direction. Therefore, the second principal component is given by

$$Z_2 = 0.544(\text{pop} - \overline{\text{pop}}) - 0.839(\text{ad} - \overline{\text{ad}}).$$

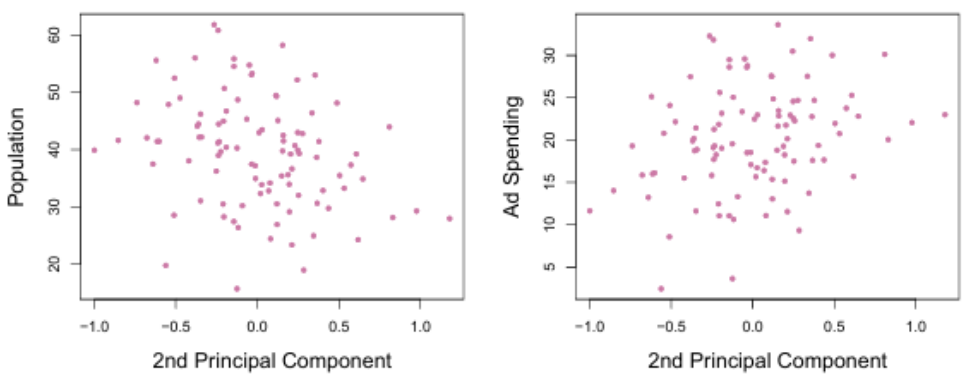


Figure 4.14: Taken from *An Introduction to Statistical Learning: Applications in Python*

By construction, the first PC contains most of the information, as seen by the larger variability in the z_{i1} and z_{i2} in the right-hand panel of Figure 4.11. There is practically no relationship between the second PC and the two predictors, as observed in Figure 4.14. This suggests we only need the information provided from the first PC in order to accurately represent the pop and ad budgets.

Principal Components Regression Approach

The *principal components regression* (PCR) approach involves constructing the first M PCs, $Z_1, \ldots, Z_M$, and then using these components as the predictors in a linear regression model that is fit using least squares. We wish to use a small number of PCs to sufficiently explain most of the variability in the data, as well as the relationship with the response. In other words, we assume that the directions in which $X_1, \ldots, X_P$ show the most variation are the directions that are associated with Y. Although this assumption is not guaranteed to hold up, it can be a reasonable enough approximation to provide promising results.

Should the PCR assumption hold, then fitting a least squares model to $Z_1, \ldots, Z_M$ will lead to better results than fitting a least squares model to $X_1, \ldots, X_p$. This is because most or all of the information are contained in M PCs and by only estimating $M << p$ coefficients we can mitigate overfitting. In the *Advertising* data set, we found that the first PC explains most of the variance between pop and ad, so a PCR that uses this single variable to predict sales will likely perform well.

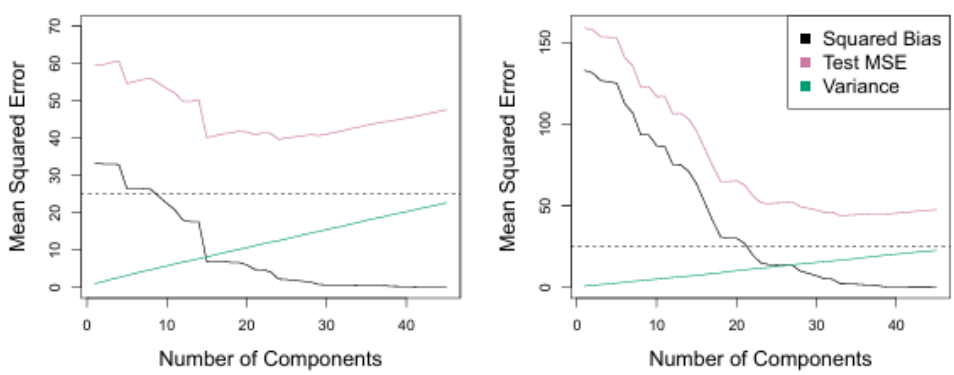


Figure 4.15: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.15 exhibits a PCR regression vs. shrinkage methods comparison for the squared bias, test MSE, and variance. They were applied to the same $p = 45, n = 50$ data as Figures 4.8 and 4.10. The irreducible error, $\text{Var}(\epsilon)$, is given by the dashed line in both plots. The left-hand panel is plotted as a function of M and the right-hand panel is plotted as a function of λ. As more PCs are used in the regression model, the bias decreases, but the variance increases, resulting in a U-shape for the MSE.

Figure 4.15 also shows how an appropriate choice for M can have a major advantage over the least squares solution. However, we can easily observe that PCR does not perform as well as the shrinkage methods. This is a consequence of the fact that the data were generated in such a way that many principal components are required in order to adequately model the response. PCR does way better in cases when the first few PCs are

sufficient to capture most of the variation in the predictors as well as the relationship with the response.

Keep in mind that PCR is not a feature selection method! Each of the M PCs used in the regression is a linear combination of all p of the original features. It does not result in the development of a model that relies upon a small set of the original features. In this sense, PCR is more closely related to ridge regression than the lasso!

The choice for M is often achieved through cross-validation. By plotting the cross-validation MSE over M, we select the value of M that achieves the minimum CV MSE.

PCR is an example of a model where predictors are recommended to be *standardized*, ensuring that all variables are on the same scale. If not, high-variance predictors will play a larger role in the PCs obtained, and the magnitude on which the variable are measured will have an effect on the final PCR model. An argument against standardization would work if each predictor follows the same unit of measurement (ex. kilograms, inches, square feet).

4.3.2 Partial Least Squares

The *linear combinations* identified in PCR are done so in an *unsupervised* way, since the response is not used to help determine the PC directions. Hence, PCR's main drawback lies in that the directions that best explain the predictors will also be the directions used for predicting the response.

Partial least squares (PLS) is a supervised alternative to PCR, and it is also a dimension reduction technique. PLS also constructs a new set of features $Z_1, \ldots, Z_M$ that are linear combinations of the original features, and then fits a linear model via least squares using these M new features. PLS, however, attempts to find directions associated with *both* the predictors and the response.

After standardizing the p predictors, PLS computes the first direction Z_1 by setting each ϕ_{j1} equal to the coefficient from the simple linear regression of Y onto X_j. Hence, in computing $Z_1 = \sum_{j=1}^{p} \phi_{j1} X_j$, PLS places the highest weight on the variables that are most strongly related to the response.

Figure 4.16 displays an example of PLS on a synthetic dataset with Sales in each of 100 regions as the response, and two predictors; Population Size and Advertising Spending. The solid green line indicates the first PLS direction, while the dotted line shows the first principal direction. PLS has chosen a solution where population is more correlated with the response than advertising compared to the PCA model. Although the PLS direction does not fit the predictors as closely as does PCA, it does a better job explaining the response.

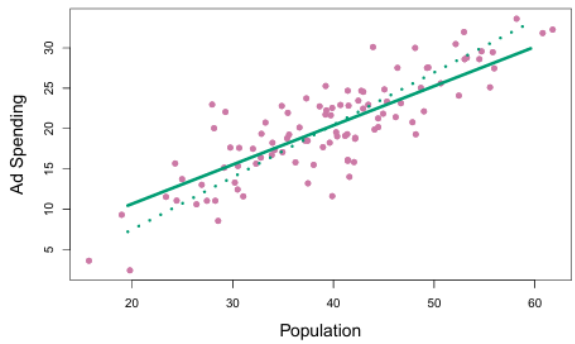


Figure 4.16: Taken from *An Introduction to Statistical Learning: Applications in Python*

Identifying the second PLS direction requires us to *adjust* each of the variables for Z_1 by regressing each variable on Z_1 and *taking residuals*. These residuals can be interpreted as the remaining information that has not been explained by the first PLS direction. Z_2 is then computed using this *orthogonalized* data in exactly the same fashion as Z_1 was computed based on the original data. Repeat M times to obtain $Z_1, \ldots, Z_M$. Then, fit a linear model using least squares to predict Y.

As with PCR, PLS uses cross-validation to select the most appropriate choice for M. In practice, PLS does not perform as well as ridge regression or PCR. Using a supervised method such as PLS can reduce bias, but it also has the potential to increase variance.

4.4 Considerations in High Dimensions

We aim to address how models can adapt or fail to adapt to the challenge of using a large number predictors.

4.4.1 High-Dimensional Data

Most traditional statistical techniques for regression and classification are intended for *low-dimensional* environments, in which $n >> p$. For example, we can develop a model to predict a patient's blood pressure on the basis of his/her age, sex, and BMI. With only four coefficients to estimate, we could have hundreds of thousands of patients, making this a low-dimensional problem.

Recently, technology has reshaped the way data is collected and analyzed, primarily in fields such as finance, marketing, and medicine. It is now very common to collect an almost unlimited number of feature measurements. Even though p is extremely large, n may be limited due to cost, sample availability, or other considerations.

For instance, rather than just predicting blood pressure on the basis mentioned earlier, one might also collect measurements for 500,000 single nucleotide polymorphisms

(SNPs: individual DNA mutations that are relatively common in the population). Then $n \approx 200$ and $p \approx 500{,}000$.

Data sets containing more features than observations are often referred to as *high-dimensional*. In such situations, using least squares is not an option. Issues in high-dimensional data often arise from the bias-variance trade-off.

4.4.2 Problems of High-Dimensional Data

To motivate the concern of high-dimensional data, we go back to the linear regression setting. Figure 4.17 shows a simple linear regression model, where the left-hand panel contains the least squares solution to $n = 20$ observations and the right-hand panel contains the least squares solution to $n = 2$ observations. With 2 observations, the line will achieve a perfect fit. However, this is problematic, and it will certainly lead to overfitting the data. This extends to when $p >> n$; there is no "unique" solutions and we can fit multiple least squares lines that will pass through all n observations. This is a result of *flexible* models.

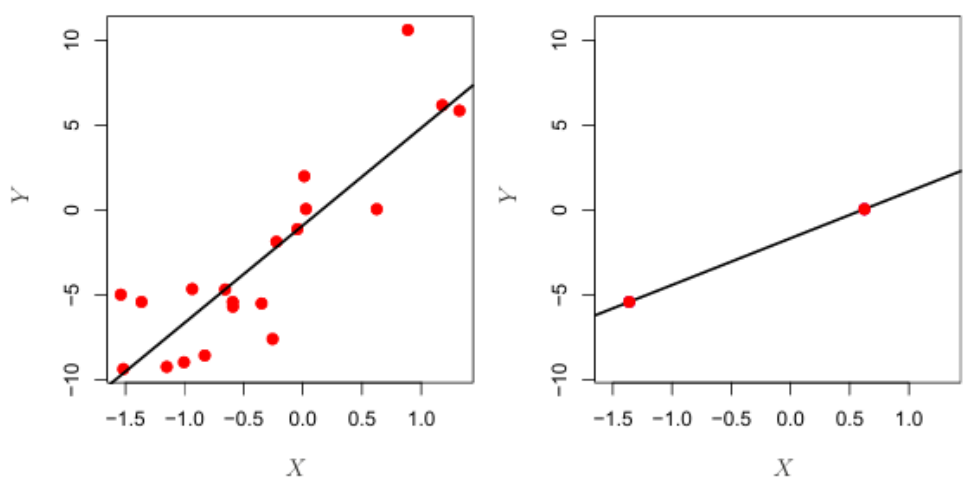


Figure 4.17: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.18 exacerbates the problem of high-dimensional data. It uses the same $n = 20$ observations from Figure 4.17; the R^2, training MSE, and test MSE are all plotted over the number of predictors used, up to $p = 20$. The R^2 progresses to 1 as p increases, thereby decreasing the training MSE to 0, *even though the features are completely unrelated to the response*. On the other hand, the MSE on an independent set grows extremely large, because including the additional predictors leads to a vast increase in variance of the coefficient estimates. In these situations, we must exercise caution and evaluate model performance on unseen test data.

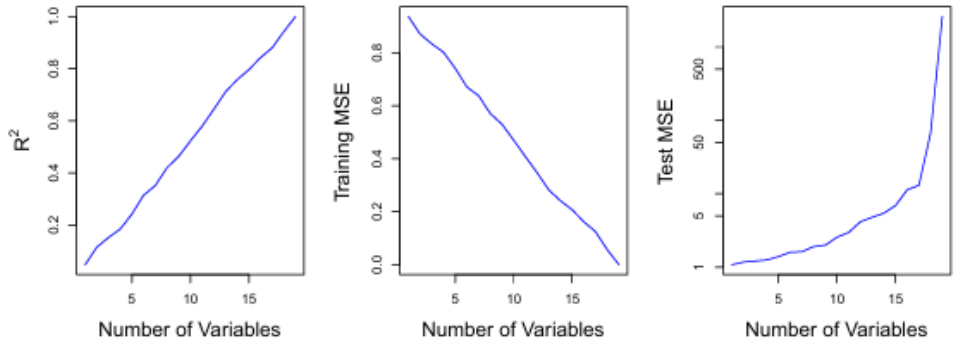


Figure 4.18: Taken from *An Introduction to Statistical Learning: Applications in Python*

Unfortunately, using approaches to adjust the training set RSS or R^2 such as C_p, AIC, and BIC are inappropriate in the high-dimensional setting, mainly because estimating $\hat{\sigma}^2$ is problematic. In cases where $p > n$, we get $\hat{\sigma}^2 = 0$ as well as $R^2 = 1$.

4.4.3 Regression in High Dimensions

It turns out that many methods in this chapter for fitting "less flexible" least squares models, such as forward stepwise selection, ridge regression, the lasso, PCR, are all useful for overcoming the limitation of standard least squares in the high-dimensional setting.

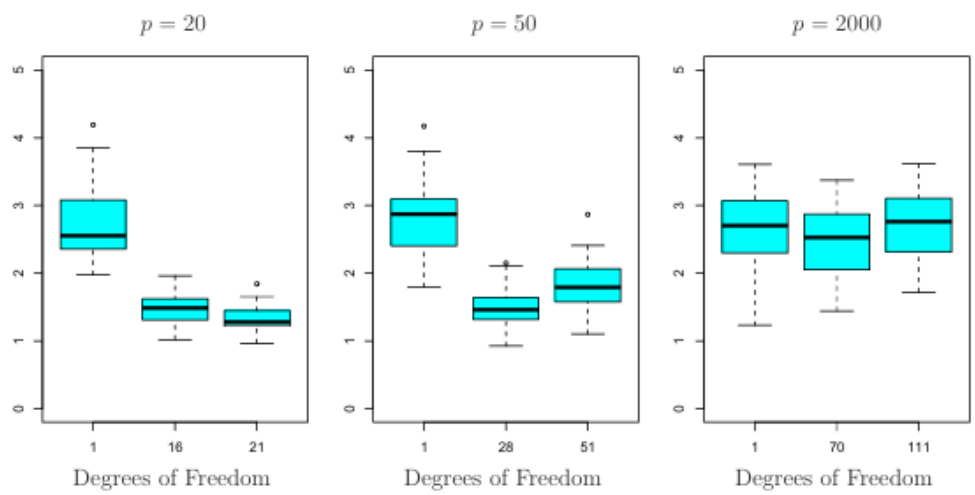


Figure 4.19: Taken from *An Introduction to Statistical Learning: Applications in Python*

Figure 4.19 illustrates the performance of the lasso in a simulated example, with $p = 20, 50$, or 2,000 features, of which 20 are truly associated with the outcome. The lasso

was performed on 100 training observations, and the MSE was evaluated on an independent test set. As the number of features increases, the test set error increases. For $p = 20$, the lowest validation set error was achieved when λ is small. As p grows, the lowest validation set error was achieved using a larger value of λ. The degrees of freedom, or number of non-zero coefficient estimates, are also shown. This is a measure of the flexibility of the lasso fit. This figure highlights some important points:

1. Regularization/shrinkage plays a key role in high-dimensional problems.
2. Appropriate tuning parameter selection is crucial for good predictive performance; and
3. The test error tends to increase as the dimensionality of the problem increases, unless the additional features are truly associated with the response.

The third point is a consequence of the curse of dimensionality, which is very prevalent in high-dimensional data. The inclusion of new predictors does not imply a better fit! The test MSE nearly doubles when p jumps from 20 to 2000. In general, adding additional signal features that are truly associated with the response will improve the fitted model (reduction in test MSE). Adding noise features would have the opposite effect: noise features increase the dimensionality of the problem, exacerbating the risk of overfitting (since noise features may be assigned nonzero coefficients). Even if features are related in high-dimensional data, the variance incurred in fitting their coefficients may outweigh the reduction in bias that they bring.

4.4.4 Interpreting Results in High Dimensions

Recall that when discussing linear regression, multicollinearity can inflate variance of the coefficient estimates and produce unreliable p-values. In high-dimensional data, the multicollinearity problem is extreme: any variable can be written as a linear combination of all of the other variables in the model. That is to say, we can never know exactly which variables (if any) truly are predictive of the outcome, and we can never identify the *best* coefficients for use in the regression. At best, we can hope that large coefficient values are assigned to predictors that are related to the response.

For example, if we used forward stepwise selection with 500,000 SNPs and obtained a model with 17 SNPs, it would be incorrect to conclude that these 17 SNPs predict blood pressure more effectively than the other SNPs not included in the model. there are likely to be many sets of 17 SNPs that would predict blood pressure just as well as the selected model. We must be careful not to overstate the results, and to instead make it clear that what we have identified is simply *one of many possible models* for predicting blood pressure, and that it must be further validated on independent data sets.

Lastly, reporting errors and measures of model fit must be carefully taken into consideration in high-dimensional data. When $p >> n$, it is easy to obtain a useless model that has zero residuals. For this reason, one should always avoid using SSE, p-values, R^2 statistics, or other traditional measures of model fit on the training data as measures of good model fit. It is important to instead report results on an independent test set, or cross-validation errors (such as MSE or R^2 on an independent test set, but training MSE is certainly not).

4.5 Examples

Example 4.4 (ISLP 6.6.1). We perform best subset, forward stepwise, and backward stepwise selection on a single data set. For each approach, we obtain $p + 1$ models, containing $0, 1, 2, \ldots, p$ predictors. Explain your answers.

(a) Which of the three models with k predictors has the smallest *training* RSS?

Solution: Best subset selection as it considers every possible model from the 2^p total. Forward and backward selection may miss the smallest RSS model.

(b) Which of the three models with k predictors has the smallest *test* RSS?

Solution: There is not enough information to select a model. A subset model with the lowest training RSS does not guarantee to yield the lowest test RSS depending on the validation method as well as the training and test sets.

(c) True or False:

i. The predictors in the k-variable model identified by forward stepwise are a subset of the predictors in the $(k + 1)$-variable model identified by forward stepwise selection.

Solution: True. The $(k + 1)$th predictor is decided by its statistics when added to the model with the k chosen predictors.

ii. The predictors in the k-variable model identified by backward stepwise are a subset of the predictors in the $(k + 1)$-variable model identified by backward stepwise selection.

Solution: True. The model with k predictors are contained in the model with $k + 1$ predictors.

iii. The predictors in the k-variable model identified by backward stepwise are a subset of the predictors in the $(k + 1)$-variable model identified by forward stepwise selection.

Solution: False.

iv. The predictors in the k-variable model identified by forward stepwise are a subset of the predictors in the $(k+1)$-variable model identified by backward stepwise selection.

Solution: False

v. The predictors in the k-variable model identified by best subset are a subset of the predictors in the $(k+1)$-variable model identified by best subset selection.

Solution: False. Predictors are neither iteratively added nor removed in best subsets, so there is no guarantee that the best model with $k+1$ predictors uses all of the k predictors.

Example 4.5 (ISLP 6.6.2). For parts (a) through (c), indicate which of i. through iv. is correct. Justify your answer.

(a) The lasso, relative to least squares, is:

i. More flexible and hence will give improved prediction accuracy when its increase in bias in less than its decrease in variance.

ii. More flexible and hence will give improved prediction accuracy when its increase in variance is less than its decrease in bias.

iii. Less flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.

iv. Less flexible and hence will give improved prediction accuracy when its increase in variance is less than its decrease in bias.

Solution: (iii) is correct. The lasso is less flexible since it has an additional constraint, which results in an increase in bias that is offset by a decrease in variance.

(b) Repeat (a) for ridge regression relative to least squares.

Solution: (iii) is correct. Ridge regression is less flexible for the same reason as the lasso even if their constraints are different.

(c) Repeat (a) for non-linear methods relative to least squares.

Solution: (ii) is correct. Non-linear models are more flexible, meaning they have lower bias and higher variance. Hence, they give better predictions when the increase in variance is less than the decrease in bias.

Example 4.6 (ISLP 6.6.3). Suppose we estimate the regression coefficients in a linear regression model by minimizing

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p}|\beta_j| \leq s$$

for a particular value of s. For parts (a) through (e), indicate which of i. through v. is correct. Justify your answer.

(a) As we increase s from 0, the training RSS will:

i. Increase initially, and then eventually start decreasing in an inverted U shape.
ii. Decrease initially, and then eventually start increasing in a U shape.
iii. Steadily increase.
iv. Steadily decrease.
v. Remain constant.

Solution: Steadily decrease. At $s = 0$, the constraint region is a dot at the origin, meaning all coefficients are zero with a training RSS equivalent to the null model. As s increase, the size of the diamond constraint region increases, therefore shifting the values of the RSS contours. Once the constraint region envelops $\hat{\beta}$, the training RSS is zero.

(b) Repeat (a) for test RSS.

Solution: Decrease initially, and then eventually start increasing in a U shape. By the bias-variance trade-off, as we increase the value of s the flexibility of the model increases, resulting in a decrease in bias and an increase in variance. For low values of s the model would have a high bias and for high values of s the model would have high variance. Hence, its lowest test RSS would be at a point in the middle that minimizes the test RSS.

(c) Repeat (a) for variance.

Solution: Steadily increase. Increasing the value of s also increases the flexibility, hence increasing the variance.

(d) Repeat (a) for (squared) bias.

Solution: Steadily decrease. By the bias-variance trade-off and results from part (c), the bias decreases when the variance increases. Thus, the squared bias decreases.

(e) Repeat (a) for the irreducible error.

Solution: Remain constant. The error term captures variability that is not included in our model (noise inherent in the data).

Example 4.7 (ISLP 6.6.4). Suppose we estimate the regression coefficients in a linear regression model by minimizing

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

for a particular value of λ. For parts (a) through (e), indicate which of i. through v. is correct. Justify your answer.

(a) As we increase s from 0, the training RSS will:

i. Increase initially, and then eventually start decreasing in an inverted U shape.
ii. Decrease initially, and then eventually start increasing in a U shape.
iii. Steadily increase.
iv. Steadily decrease.
v. Remain constant.

Solution: Steadily increase. At $\lambda = 0$ the coefficient estimates are equal to the least squares estimates. As λ increases, the flexibility of the model decreases, resulting in a greater training RSS until $\lambda = \infty$ (in which it equals the null model training RSS).

(b) Repeat (a) for test RSS.

Solution: Decrease initially, then eventually start increasing in a U shape. At small values of λ where the flexibility is high, the model has high variance and low bias. As λ increases, the flexibility of the model decreases, reducing the variance and increasing the bias which results in a lower value of the RSS. Eventually, the flexibility gets too low and the model becomes too rigid to fit the data due to the high bias and low variance, resulting in an increase in the test RSS.

(c) Repeat (a) for variance.

Solution: Steadily decrease. As λ increases the flexibility of the model decreases, resulting in a decrease in variance.

(d) Repeat (a) for (squared) bias.

Solution: Steadily increase. As λ increases the flexibility of the model decreases, resulting in an increase in bias.

(e) Repeat (a) for the irreducible error.

Solution: Remain constant for the same reason mentioned in 4.6(e).

Example 4.8 (SOA Practice Problem 8). Determine which of the following statements describe the advantages of using a subset selection process, such as backward selection.

- Doing so will likely result in a simpler model.
- Doing so will likely improve prediction accuracy.
- The results are likely to be easier to interpret.

(a) Statement 1 is true.

(b) Statement 2 is true.

(c) Statement 3 is true.

(d) All three statements are true.

(e) The correct answer is not given by (A), (B), (C), or (D).

Solution: D. Irrelevant predictors are excluded from the final model, resulting in a simpler and easier to interpret model. Additionally, the variance is decreased and increases the prediction accuracy. Hence, all three statements hold.

Example 4.9 (SOA Practice Problem 61). A linear model has been fit to a dataset containing six predictor variables, F, G, H, I, J, and K. Determine which of the following statements regarding Akaike information criterion (AIC) or Bayesian information (BIC) to select an optimal set of predictor variables for this linear model is/are true.

1. AIC and BIC provide a direct estimate of the test error.
2. When choosing between the subsets $\{F, G, H\}$ and $\{I, J, K\}$, AIC will always select the same subset as BIC.
3. For large sample sizes ($n > 7$), the number of variables selected by BIC will be less than or equal to the number selected by AIC.

(a) None

(b) 1 and 2 only

(c) 1 and 3 only

(d) 2 and 3 only

(e) The correct answer is not given by (A), (B), (C), or (D).

Solution: D. Statement 1 is false; AIC and BIC make an indirect estimate by adjusting the training error. Statement 2 is true; for a fixed value of the number of predictors, the two provide the same ranking. Statement 3 is true; for $n > 7$, BIC provides a greater penalty for additional variables, and hence will select a number less than equal to that selected by AIC.

Example 4.10 (SOA Practice Problem 70). A dataset has 100 observations with six predictor variables, X_1, X_2, X_3, X_4, X_5, and X_6 and a response variable, Y. Three ordinary regression models have been run, with the following results.

Model Number	Variables Used	RSS	RSE
I	X_1 only	183,663.30	43.2911
II	X_1 and X_2 only	8,826.47	9.5391
III	All 6 predictors	8,319.59	9.4582

The Akaike Information Criterion (AIC) is to be used to select the best method from these three choices.

Determine which of the following statements is true.

(a) Select Model I with an AIC value of 1,838.42

(b) Select Model I with an AIC value of 1,874.12

(c) Select Model II with an AIC value of 91.84

(d) Select Model II with an AIC value of 91.90

(e) Select Model III with an AIC value of 93.93

Solution: C. Including X_2 into the model decreases the RSS substantially, suggesting that when X_1 is included in the model, X_2 is associated with the response. The RSS decreases slightly with all six predictors in Model III; however, since Model II has the lowest AIC (91.84), we should select Model II. This decision should be made not only from the AIC, but also because X_1 and X_2 alone explain a lot of the variability with the response, hence making the model easier to interpret.

5 Time Series: Part I